

COS20019 - Cloud Computing Architecture

Week 8: ACA Module 9 Guided Lab - Creating a Highly Available Environment

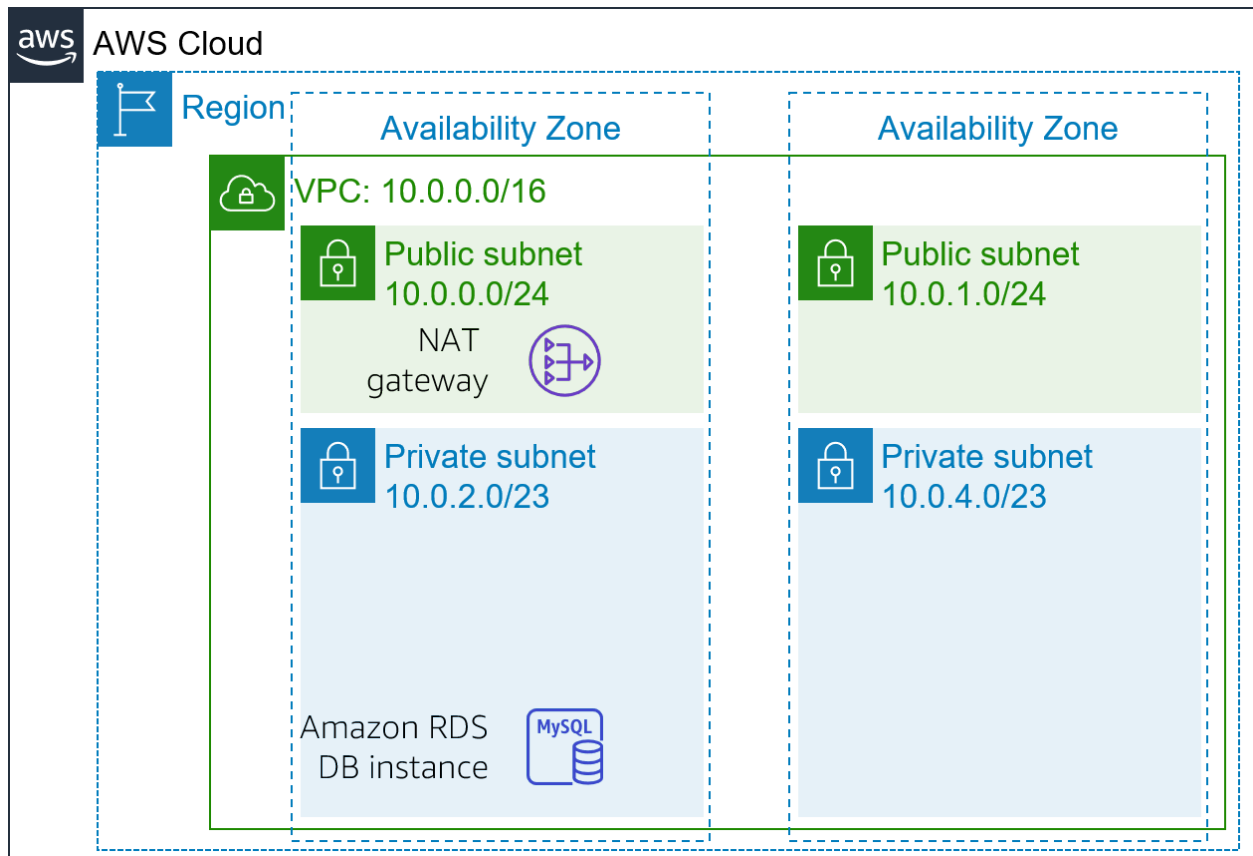
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Due Date: 29/10/2023

Task 1: Inspecting your VPC

This lab begins with an environment that is already deployed via AWS CloudFormation. It includes:

- A VPC
- Public and private subnets in two Availability Zones
- An internet gateway (not shown) that is associated with the public subnets
- A Network Address Translation (NAT) gateway in one of the public subnets
- An Amazon Relational Database Service (Amazon RDS) instance in one of the private subnets



In this task, you will review the configuration of the VPC that was created for this lab.

5. On the **AWS Management Console**, on the Services menu, choose **VPC**.
6. In the left navigation pane, under **Filter by VPC**, choose the **Select a VPC** box and select **Lab VPC**.
This setting will limit the console to only show resources that are associated with the *Lab VPC*.

7. In the left navigation pane, choose **Your VPCs**.

Here, you can access information about the **Lab VPC** that was created for you. Choose **Lab VPC**.

In the **Details** tab, notice that the **IPv4 CIDR** field has a value of *10.0.0.0/16*, which means that this VPC includes all IP addresses that start with *10.0.x.x*.

The screenshot shows the AWS VPC Console interface. The left navigation pane is expanded, showing 'Your VPCs' under the 'Virtual private cloud' section. The main content area displays a table of VPCs with one entry: 'Lab VPC' with VPC ID 'vpc-045a22d2a5e361490' and state 'Available'. Below the table, the 'Details' tab is selected for the 'Lab VPC'. The details are organized into a grid:

vpc-045a22d2a5e361490 / Lab VPC			
Details			
VPC ID vpc-045a22d2a5e361490	State Available	DNS hostnames Enabled	DNS resolution Enabled
Tenancy Default	DHCP option set dopt-04d6887fd28815c37	Main route table rtb-0b41a008a6b3839ca	Main network ACL acl-0594fb419ab20e419
Default VPC No	IPv4 CIDR 10.0.0.0/16	IPv6 pool -	IPv6 CIDR (Network border group) -
Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID 280813733559	

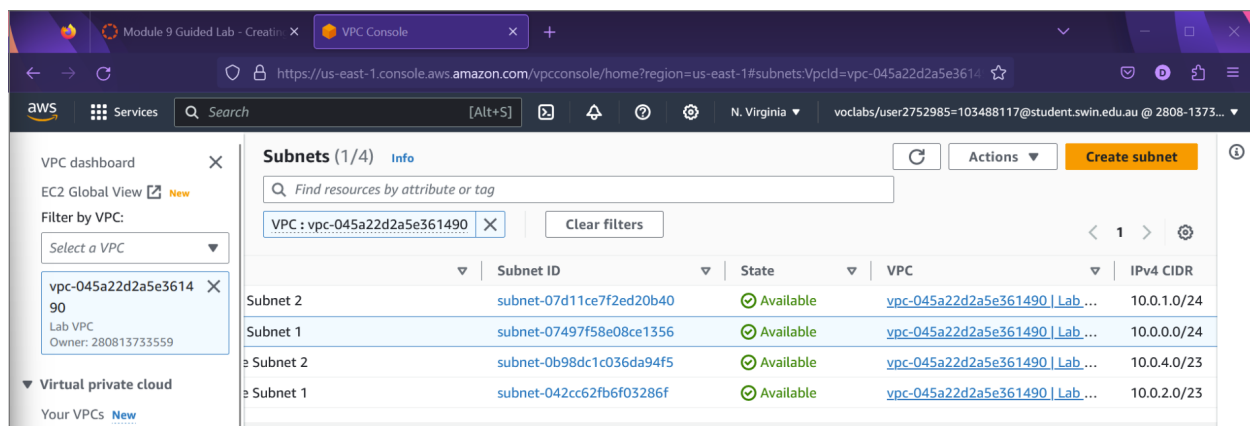
8. In the left navigation pane, choose **Subnets**.

Here, you can access information about **Public Subnet 1**. In the list of subnets view:

- The **VPC** column for Public Subnet 1 shows that this subnet exists inside of *Lab VPC*.
- The **IPv4 CIDR** column has a value of *10.0.0.0/24*, which means that this subnet includes the 256 IP addresses between *10.0.0.0* and *10.0.0.255*. Five of these addresses are reserved and unusable.
- The **Availability Zone** column lists the Availability Zone where this subnet resides.

9. To reveal more details select **Public Subnet 1**.

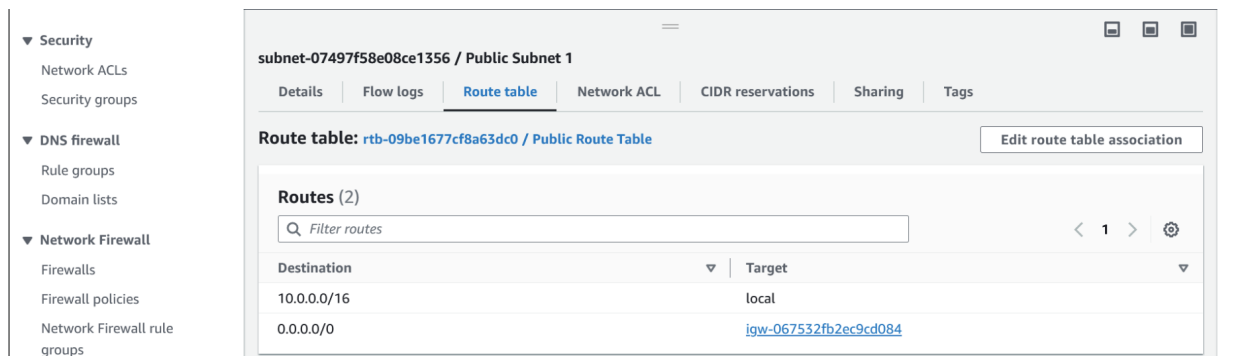
Tip: To adjust the size of the lower window pane, you can drag the divider.



10. In the lower half of the page, choose the **Route table** tab.

This tab includes details about the routing for this subnet:

- The first entry specifies that traffic destined within the Classless Inter-Domain Routing (CIDR) range for the VPC (*10.0.0.0/16*) will be routed within the VPC (*local*).
- The second entry specifies that any traffic destined for the internet (*0.0.0.0/0*) is routed to the internet gateway (*igw-*) that exists in Lab VPC. This setting makes the subnet a *public subnet*.



11. Choose the **Network ACL** tab.

This tab has information about the network access control list (network ACL) that is associated

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with the subnet. The rules are currently set to the default settings. They permit *all traffic* to flow in and out of the subnet, but the rules can be further restricted by using security groups.

The screenshot shows the AWS VPC console interface. The left-hand navigation pane is open, displaying the 'VPC dashboard' and a list of VPCs. The selected VPC is 'vpc-045a22d2a5e361490', labeled 'Lab VPC' with owner '280813733559'. The main content area is titled 'Network ACLs (1/1)' and shows a table with one entry:

Name	Network ACL ID	Associated with	Default	VPC ID
-	acl-0594fb419ab20e419	4 Subnets	Yes	vpc-045a22d2a5e361490

12. In the left navigation pane, choose **Internet gateways**.

Notice that an internet gateway with the name **Lab IG** is already attached to **Lab VPC**.

The screenshot shows the AWS VPC console interface with the 'Internet gateways' section selected in the left navigation pane. The main content area is titled 'Internet gateways (1/1)' and shows a table with one entry:

Name	Internet gateway ID	State	VPC ID
Lab IG	igw-067532fb2ec9cd084	Attached	vpc-045a22d2a5e361490

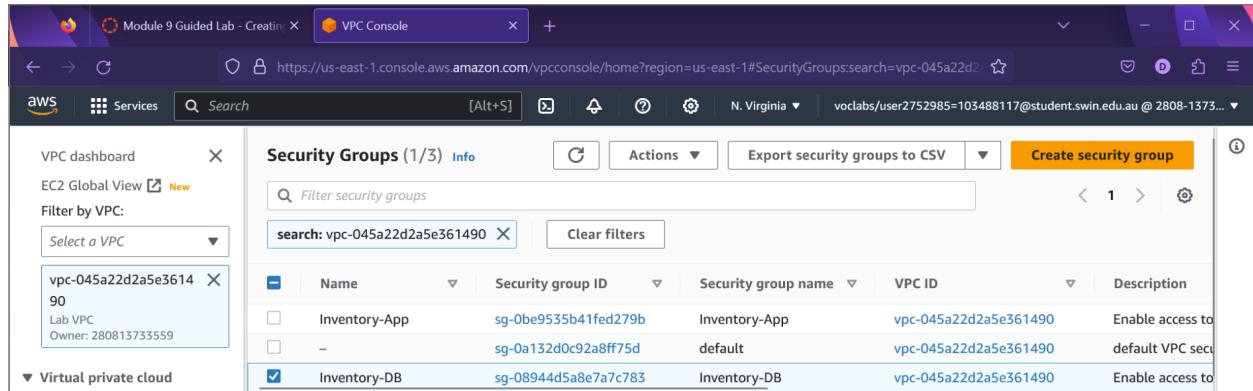
Below the table, the details for the selected internet gateway 'igw-067532fb2ec9cd084 / Lab IG' are displayed:

Details			
Internet gateway ID	State	VPC ID	Owner
igw-067532fb2ec9cd084	Attached	vpc-045a22d2a5e361490 Lab VPC	280813733559

13. In the left navigation pane, choose **Security groups**.

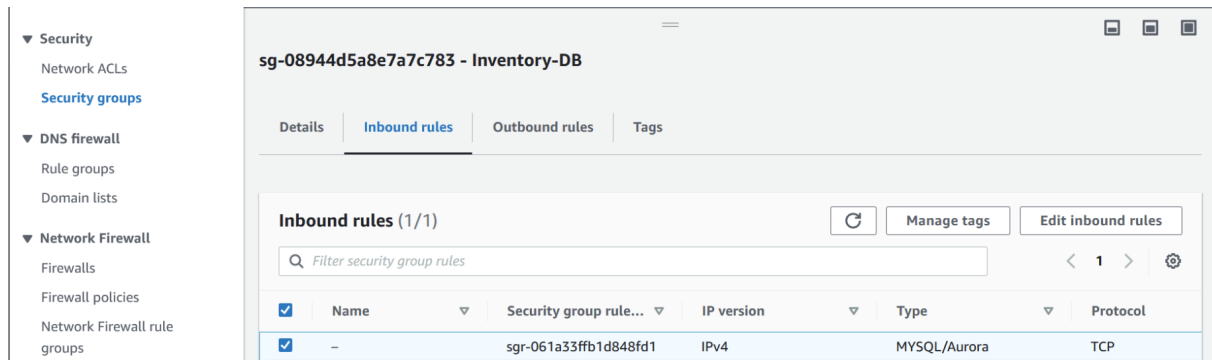
14. Select **Inventory-DB**.

This security group controls incoming traffic to the database.



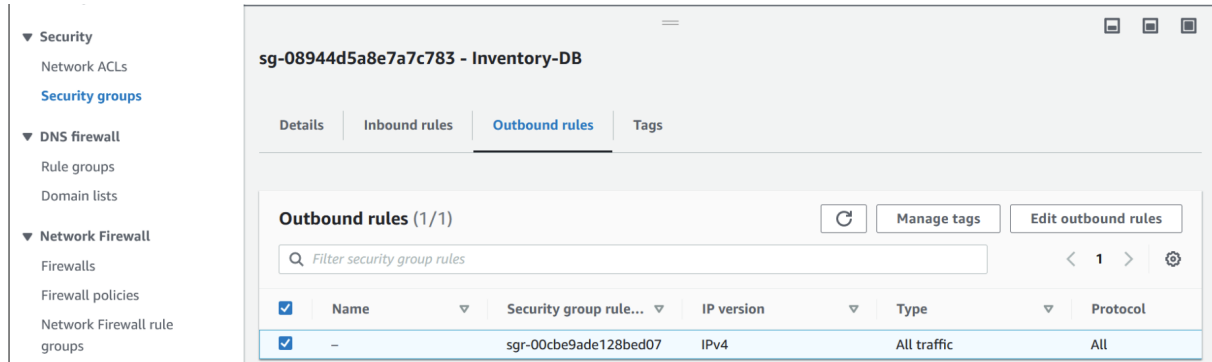
15. In the lower half of the page, choose the **Inbound rules** tab.

The rule defined in this security group permits inbound MySQL or Aurora traffic (port 3306) from anywhere in the VPC (*10.0.0.0/16*). You will later modify this setting so it only accepts traffic from the application servers.



16. Choose the **Outbound rules** tab.

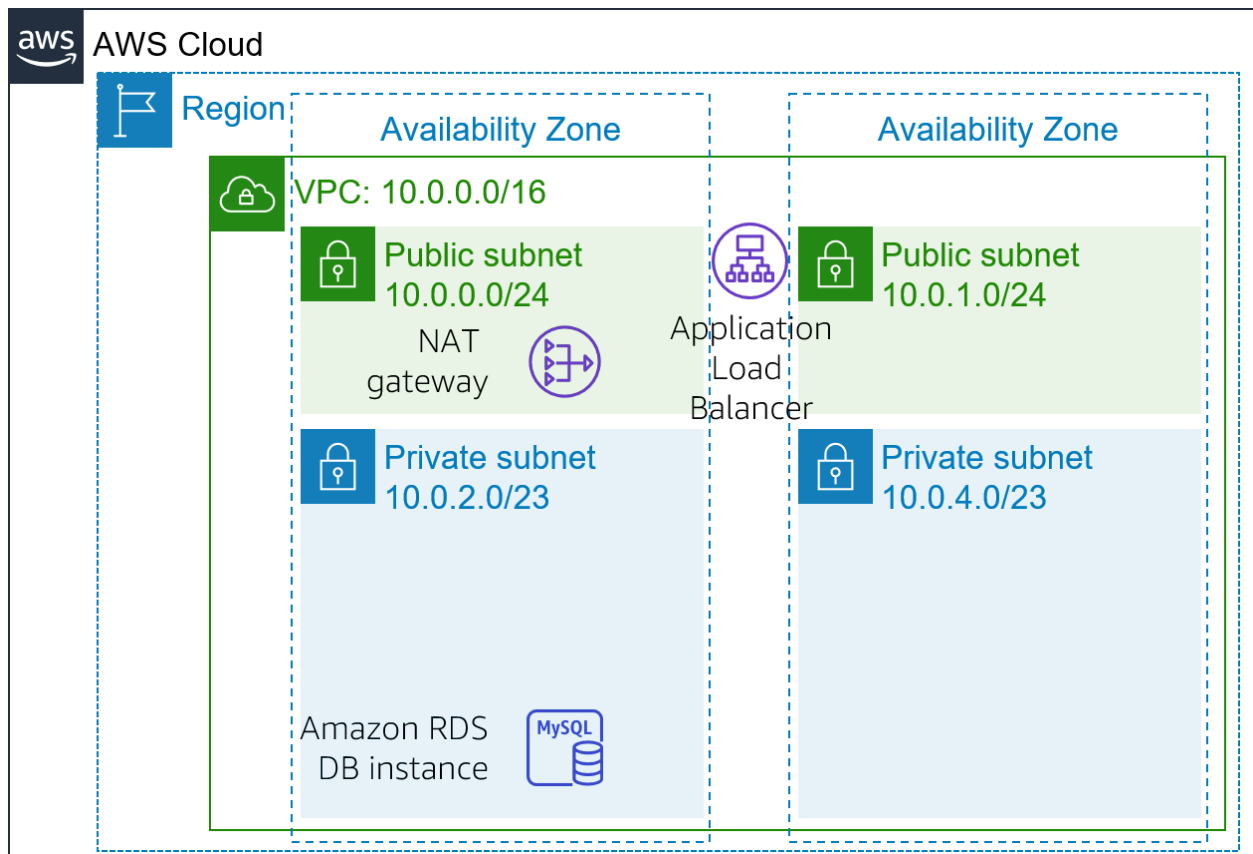
By default, security groups allow all outbound traffic. However, these settings can be modified as needed.



Task 2: Creating an Application Load Balancer

To build a highly available application, it is a best practice to launch resources in *multiple Availability Zones*. Availability Zones are physically separate data centers (or groups of data centers) in the same Region. If you run your applications across multiple Availability Zones, you can provide greater *availability* if a data center experiences a failure.

Because the application runs on multiple application servers, you will need a way to distribute traffic amongst those servers. You can accomplish this goal by using a *load balancer*. This load balancer will also perform health checks on instances and only send requests to healthy instances.



17. On the Services menu, choose **EC2**.
18. In the left navigation pane, choose **Load Balancers** (you might need to scroll down to find it).
19. Choose Create load balancer
Several types of load balancers are displayed. Read the descriptions of each type to understand their capabilities.
20. Under **Application Load Balancer**, choose **Create**.
21. For **Load balancer name**, enter: **Inventory-LB**
22. Scroll down to the **Network mapping** section, then for **VPC**, select **Lab VPC**.
Important: Be sure to choose **Lab VPC**. It is likely not the default selection.
You will now specify which *subnets* the load balancer should use. It will be a public load balancer, so you will select both public subnets.
23. Under **Mappings**, choose the **first** Availability Zone, then choose the **Public Subnet** that displays.
24. Choose the **second** Availability Zone, then choose the **Public Subnet** that displays.
You should now have selected two subnets: **Public Subnet 1** and **Public Subnet 2**. (If not, go back and try the configuration again.)

Basic configuration

Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)
Scheme can't be changed after the load balancer is created.
☒ **Internet-facing**
An internet-facing load balancer routes requests from clients over the internet to targets. Requires a public subnet. [Learn more](#)
☐ **Internal**
An internal load balancer routes requests from clients to targets using private IP addresses.

IP address type [Info](#)
Select the type of IP addresses that your subnets use.
☒ **IPv4**
Recommended for internal load balancers.
☐ **Dualstack**
Includes IPv4 and IPv6 addresses.

Network mapping [Info](#)
The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC [Info](#)
Select the virtual private cloud (VPC) for your targets or you can [create a new VPC](#). Only VPCs with an internet gateway are enabled for selection. The selected VPC can't be changed after the load balancer is created. To confirm the VPC for your targets, view your [target groups](#).

vpc-045a22d2a5e361490
IPv4: 10.0.0.0/16

Mappings [Info](#)
Select at least two Availability Zones and one subnet per zone. The load balancer routes traffic to targets in these Availability Zones only. Availability Zones that are not supported by the load balancer or the VPC are not available for selection.

☒ **us-east-1a (use1-az4)**
Subnet
 Public Subnet 1
IPv4 address
Assigned by AWS

☒ **us-east-1b (use1-az6)**
Subnet
 Public Subnet 2
IPv4 address
Assigned by AWS

25. In the **Security groups** section, select the **Create a new security group** hyperlink. This opens a new browser tab. Configure the new security group settings:

- **Security group name:** **Inventory-LB**
- **Description:** **Enable web access to load balancer**
- **VPC:** Remove the default VPC by choosing the X to the right of it. Then select **Lab VPC**.

26. Under **Inbound rules**, choose **Add rule** and configure as described:

- **Type:** **HTTP**

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- **Source:** *Anywhere-IPv4*

27. Still under **Inbound rules**, choose **Add rule** again and configure:

- **Type:** *HTTPS*
- **Source:** *Anywhere-IPv4*

EC2 > Security Groups > Create security group

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

Name cannot be edited after creation.

Description [Info](#)

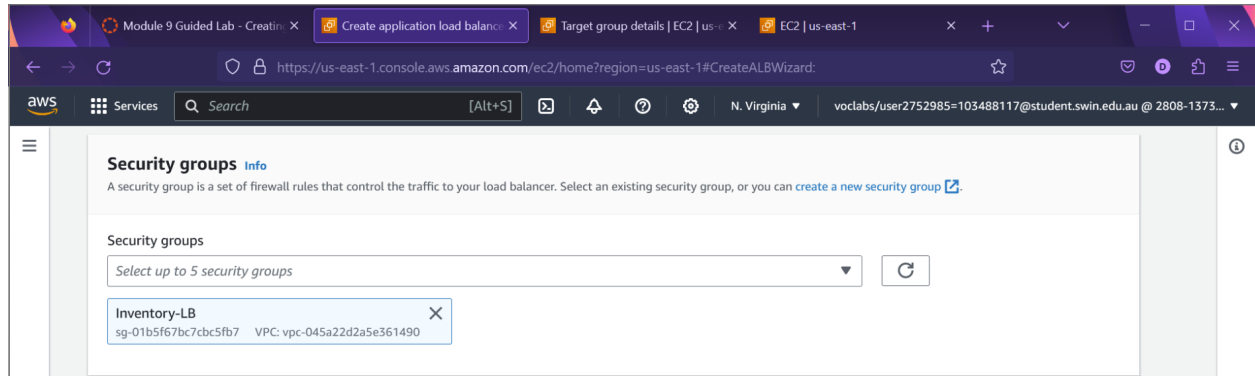
VPC [Info](#)

Inbound rules [Info](#)

28. Choose **Create security group**.

29. Assign the security group to the load balancer:

- Return to the browser tab where you are still configuring the load balancer.
- In the **Security groups** area and choose the refresh icon.
- For **Security groups**, select the **Inventory-LB** security group you just created.
- Next, below the **Security groups** dropdown menu, select the **X** next to the *default* security group so that Inventory-LB is now the only security group chosen.



30. In the **Listeners and routing** section, choose **Create target group**.

A new browser tab will open.

Analysis: *Target groups* define where to *send* traffic that comes into the load balancer. The Application Load Balancer can send traffic to multiple target groups based upon the URL of the incoming request, such as having requests from mobile apps going to a different set of servers. Your web application will use only one target group.

31. Configure the target group as described here:

- **Choose a target type:** *Instances*
- **Target group name:** **Inventory-App**
- **VPC:** Ensure that *Lab VPC* is chosen.
- Scroll down and expand **Advanced health check settings**.
Note: The Application Load Balancer automatically performs *health checks* on all instances to ensure that they are responding to requests. The default settings are recommended, but you will make them slightly faster for use in this lab.
- **Healthy threshold:** **2**
- **Interval:** **10** (seconds)
The configurations you have chosen will result in the health check being performed every 10 seconds, and if the instance responds correctly twice in a row, it will be considered healthy.
- Choose **Next**. The *Register targets* screen appears.
Note: *Targets* are the individual instances that will respond to requests from the load balancer.
You do not have any web application instances yet, so you can skip this step.
- Review the settings and choose **Create target group**.

The screenshot shows the AWS Management Console interface for creating a target group. The browser tabs include 'Module 9 Guided Lab - Creatin...', 'Create application load balanc...', 'Step 1 Create target group | EC...', and 'EC2 | us-east-1'. The address bar shows the URL: `https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateTargetGroup:protocol=HTTP;vpc=vpc-`. The page title is 'Specify group details' with a subtitle: 'Your load balancer routes requests to the targets in a target group and performs health checks on the targets.'

Basic configuration
Settings in this section can't be changed after the target group is created.

Choose a target type

- ☒ **Instances**
 - Supports load balancing to instances within a specific VPC.
 - Facilitates the use of [Amazon EC2 Auto Scaling](#) to manage and scale your EC2 capacity.
- ☐ **IP addresses**
 - Supports load balancing to VPC and on-premises resources.
 - Facilitates routing to multiple IP addresses and network interfaces on the same instance.
 - Offers flexibility with microservice based architectures, simplifying inter-application communication.
 - Supports IPv6 targets, enabling end-to-end IPv6 communication, and IPv4-to-IPv6 NAT.
- ☐ **Lambda function**
 - Facilitates routing to a single Lambda function.
 - Accessible to Application Load Balancers only.
- ☐ **Application Load Balancer**
 - Offers the flexibility for a Network Load Balancer to accept and route TCP requests within a specific VPC.
 - Facilitates using static IP addresses and PrivateLink with an Application Load Balancer.

Target group name
Inventory-App
A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Protocol : Port
HTTP 80
1-65535

IP address type
Only targets with the indicated IP address type can be registered to this target group.

- ☒ **IPv4**
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.
- ☐ **IPv6**
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

Lab VPC
vpc-045a22d2a5e361490
IPv4: 10.0.0.0/16

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Health checks
The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol
HTTP

Health check path
Use the default path of "/" to ping the root, or specify a custom path if preferred.
/
Up to 1024 characters allowed.

Advanced health check settings
[Restore defaults](#)

Health check port
The port the load balancer uses when performing health checks on targets. By default, the health check port is the same as the target group's traffic port. However, you can specify a different port as an override.
☒ Traffic port
☐ Override

Healthy threshold
The number of consecutive health checks successes required before considering an unhealthy target healthy.
2
2-10

Unhealthy threshold
The number of consecutive health check failures required before considering a target unhealthy.
2
2-10

Timeout
The amount of time, in seconds, during which no response means a failed health check.
5 seconds
2-120

Interval
The approximate amount of time between health checks of an individual target
10 seconds
5-300

Success codes
The HTTP codes to use when checking for a successful response from a target. You can specify multiple values (for example, "200,202") or a range of values (for example, "200-299").
200

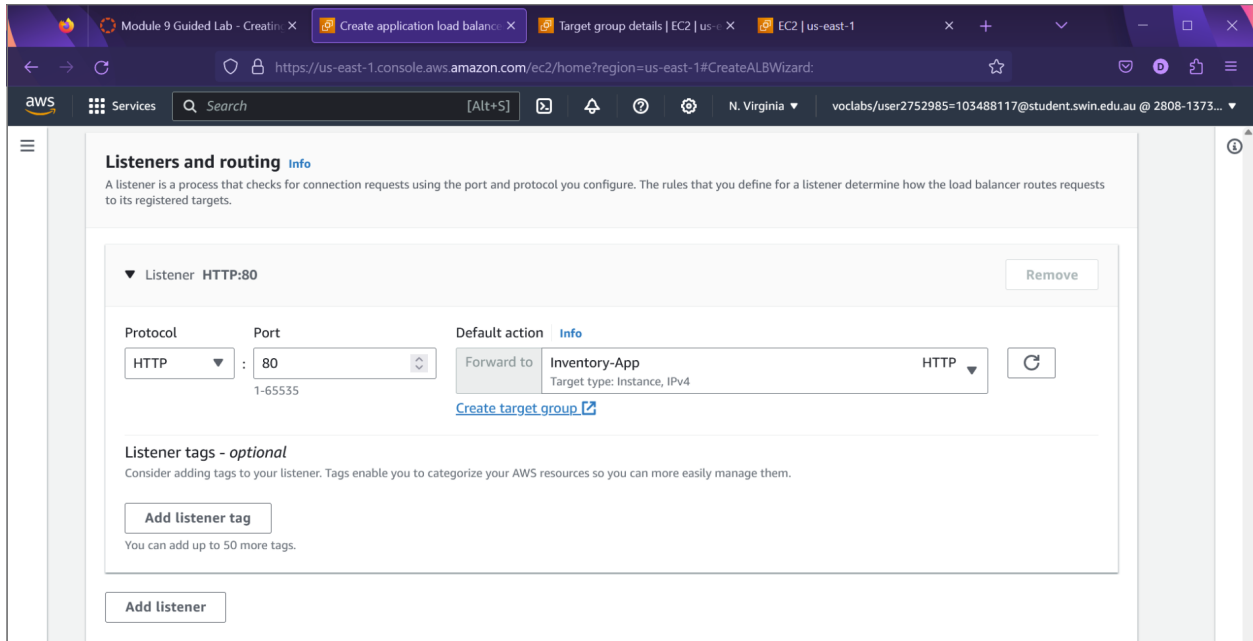
Attributes
[i](#) Certain default attributes will be applied to your target group. You can view and edit them after creating the

32. Return to the browser tab where you already started defining the load balancer.

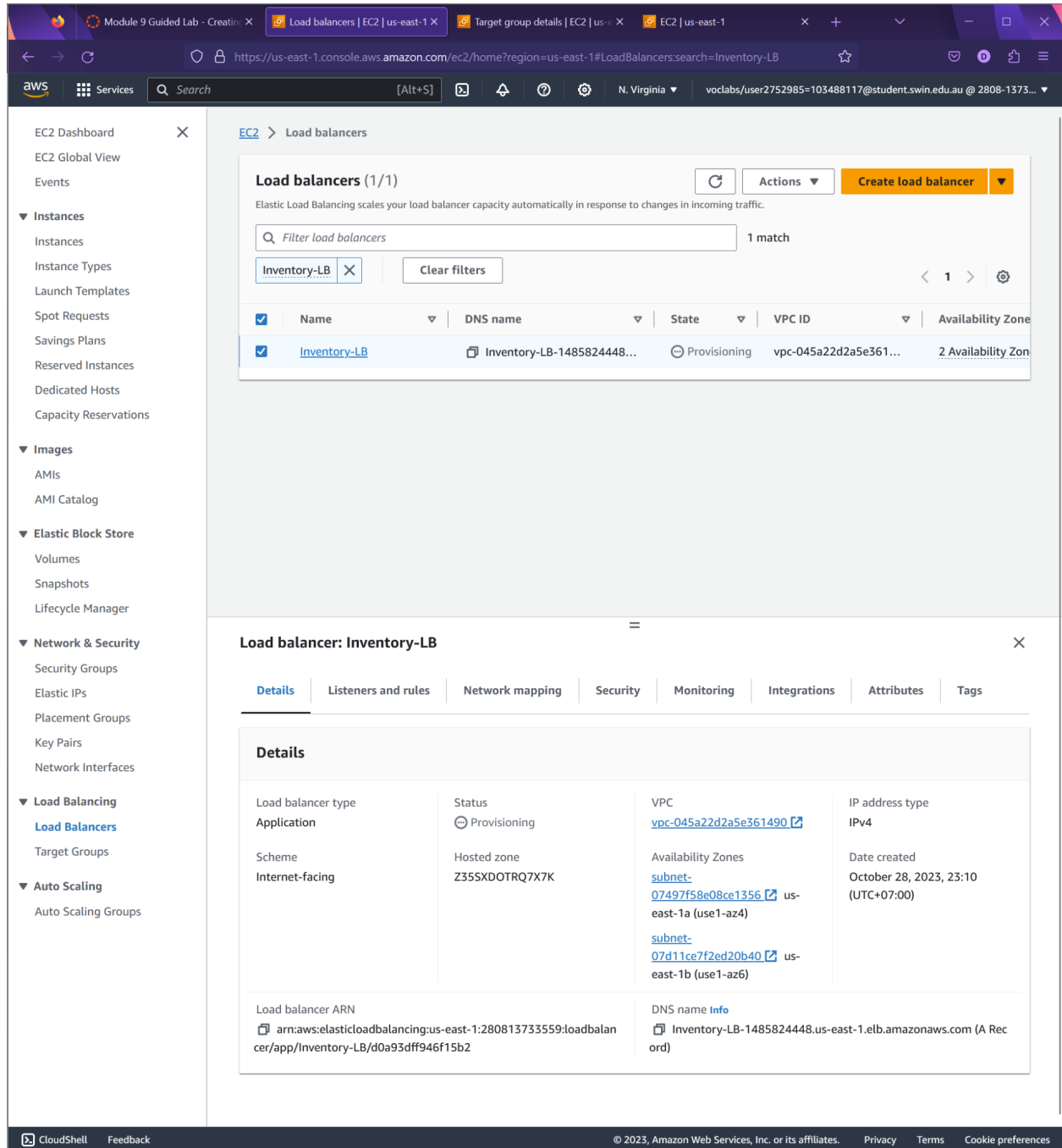
33. In the **Listeners and routing** section, choose the refresh icon.

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34. For the **Listener HTTP:80** row, set the **Default action** to forward to the **Inventory-App** target group you just created.



35. Scroll to the bottom of the page, and choose **Create load balancer**.
- The load balancer is successfully created.
 - Choose **View load balancer**.



The screenshot shows the AWS Management Console interface for the 'Load balancers' page. The left sidebar contains navigation links for various AWS services. The main content area displays a list of load balancers, with 'Inventory-LB' selected. Below the list, the details for 'Inventory-LB' are shown, including its type (Application), status (Provisioning), VPC ID, and availability zones.

Load balancers (1/1)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers: 1 match

Name	DNS name	State	VPC ID	Availability Zone
Inventory-LB	Inventory-LB-1485824448...	Provisioning	vpc-045a22d2a5e361...	2 Availability Zone

Load balancer: Inventory-LB

Details | Listeners and rules | Network mapping | Security | Monitoring | Integrations | Attributes | Tags

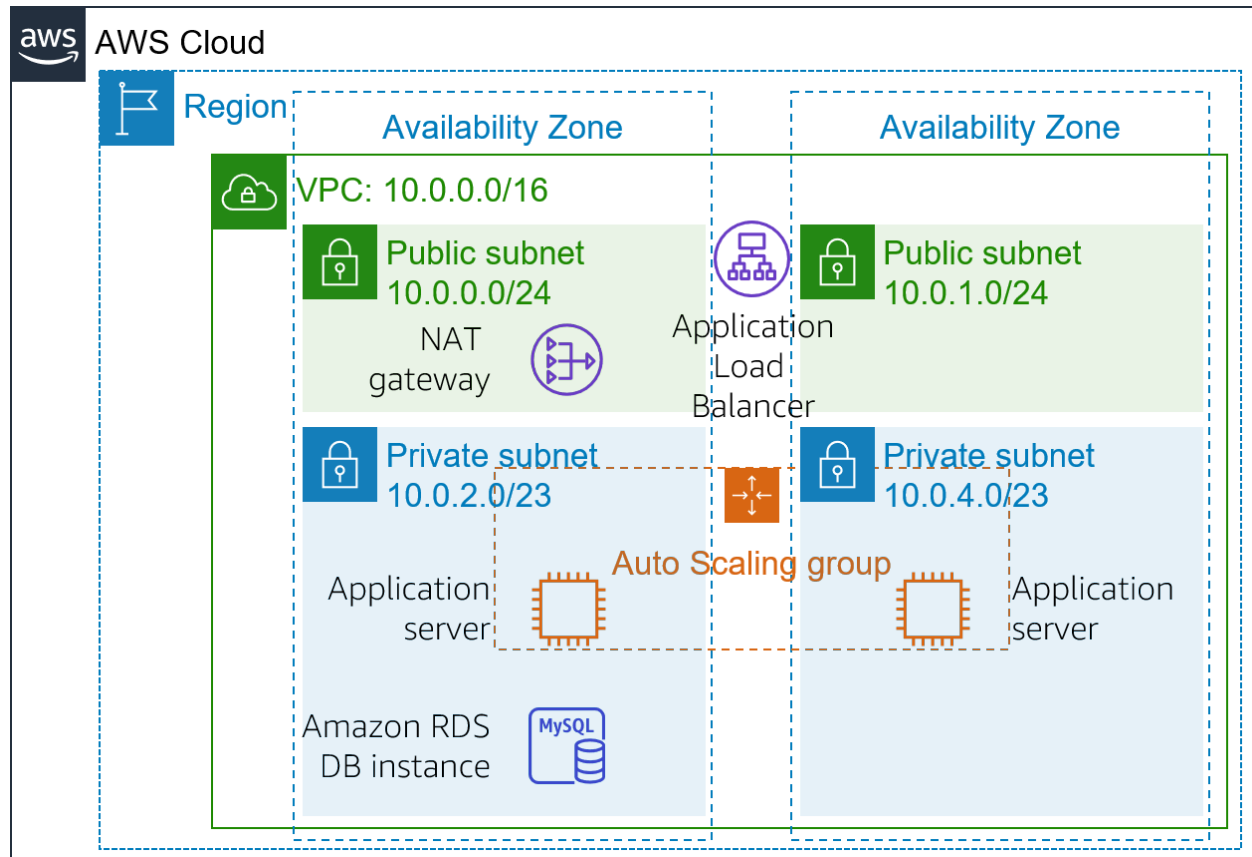
Details

Load balancer type Application	Status Provisioning	VPC vpc-045a22d2a5e361490	IP address type IPv4
Scheme Internet-facing	Hosted zone Z35SXDOTRQ7X7K	Availability Zones subnet-07497f58e08ce1356 us-east-1a (use1-az4) subnet-07d11ce7f2ed20b40 us-east-1b (use1-az6)	Date created October 28, 2023, 23:10 (UTC+07:00)
Load balancer ARN arn:aws:elasticloadbalancing:us-east-1:280813733559:loadbalancer/app/Inventory-LB/d0a93dff946f15b2		DNS name Inventory-LB-1485824448.us-east-1.elb.amazonaws.com (A Record)	

Task 3: Creating an Auto Scaling group

Amazon EC2 Auto Scaling is a service designed to *launch* or *terminate* Amazon EC2 instances automatically based on user-defined policies, schedules, and health checks. It also *automatically distributes instances across multiple Availability Zones* to make applications *highly available*.

In this task, you will create an Auto Scaling group that deploys EC2 instances across your *private subnets*, which is a security best practice for application deployment. Instances in a private subnet cannot be accessed from the internet. Instead, users send requests to the load balancer, which forwards the requests to EC2 instances in the private subnets.



Create an AMI for Auto Scaling

You will create an Amazon Machine Image (AMI) from the existing *Web Server 1*. This will save the contents of the root volume of the web server so that new instances can be launched with an identically configured guest operating system.

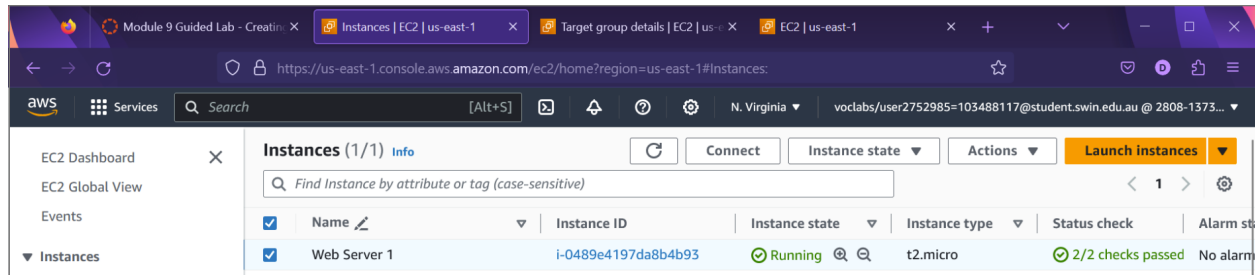
36. In the **AWS Management Console**, on the Services menu, choose **EC2**.
37. In the left navigation pane, choose **Instances**.
First, you will confirm that the instance created for you in this lab is running.

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38. Wait until the **Status check** for **Web Server 1** displays *2/2 checks passed*. Choose refresh to update.

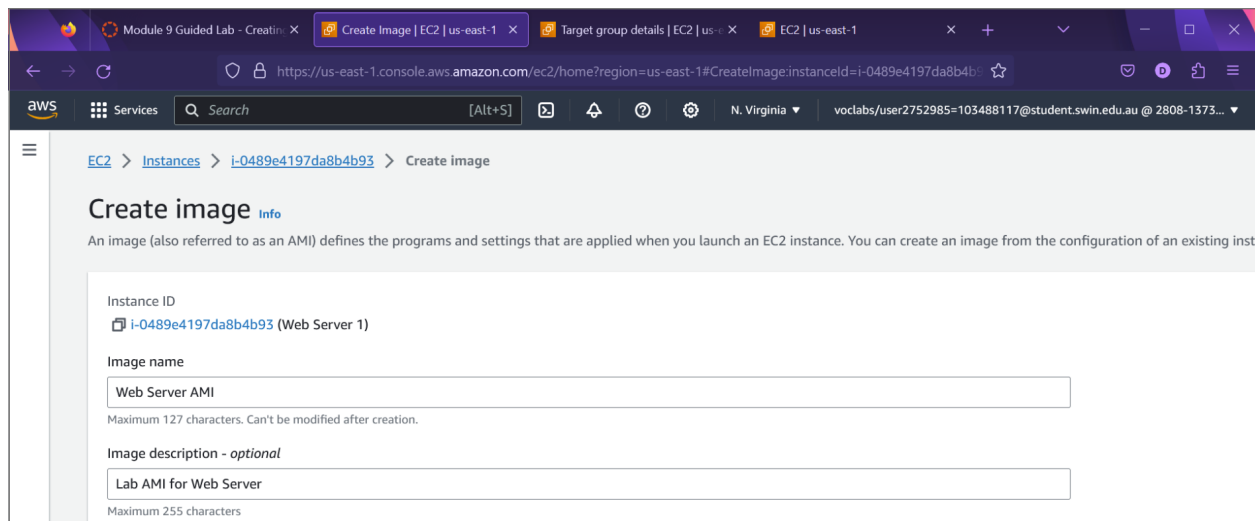
You will now create an AMI based upon this instance.

39. Select **Web Server 1**.



40. In the Actions menu, choose **Image and templates** > **Create image**, then configure:

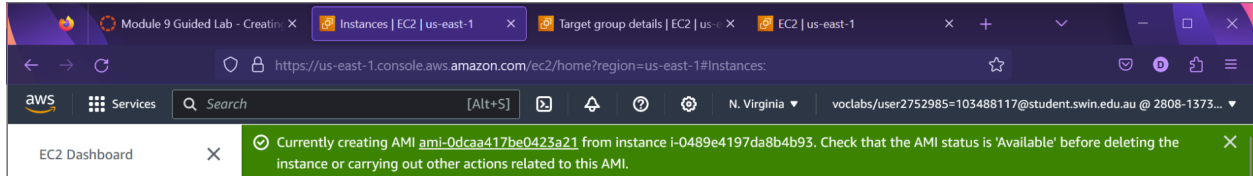
- **Image name:** Web Server AMI
- **Image description:** Lab AMI for Web Server



41. Choose **Create image**

A banner at the top of the screen displays the **AMI ID** for your new AMI.

You will use this AMI when launching the Auto Scaling group later in the lab.



Create a Launch Template and an Auto Scaling Group

You will first create a *launch template*. A launch template is a template that an Auto Scaling group uses to launch EC2 instances. When you create a launch template, you specify information for the instances such as the AMI, the instance type, a key pair, and security group.

42. In the left navigation pane, choose **Launch Templates**.

43. Choose Create launch template

44. Configure the launch template settings and create it:

- **Launch template name:** **Inventory-LT**
- Under **Auto Scaling guidance**, select *Provide guidance to help me set up a template that I can use with EC2 Auto Scaling*
- In the Application and OS Images (Amazon Machine Image) area, choose *My AMIs*.
- **Amazon Machine Image (AMI):** choose *Web Server AMI*
- **Instance type:** choose *t2.micro*
- **Key pair name:** choose *vockey*
- **Firewall (security groups):** choose *Select existing security group*
- **Security groups:** choose *Inventory-App*
- Scroll down to the **Advanced details** area and expand it.
 - **IAM instance profile:** choose *Inventory-App-Role*
 - Scroll down to the **Detailed CloudWatch monitoring** setting. Select *Enable*
Note: This will allow Auto Scaling to react quickly to changing utilization.
 - Under **User data**, paste in the script below:
 - ```
#!/bin/bash
Install Apache Web Server and PHP
yum install -y httpd mysql
amazon-linux-extras install -y php7.2
Download Lab files
wget
https://aws-tc-largeobjects.s3-us-west-2.amazonaws.com/ILT-TF-200-ACACAD-20-EN/mod9-guided/scripts/inventory-app.zip
unzip inventory-app.zip -d /var/www/html/
Download and install the AWS SDK for PHP
```

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```
wget https://github.com/aws/aws-sdk-php/releases/download/3.62.3/aws.zip
unzip aws -d /var/www/html
Turn on web server
chkconfig httpd on
service httpd start
```

- Choose Create launch template

Next, you will create an Auto Scaling group that uses this launch template. The *Auto Scaling group* defines *where to launch* the EC2 instances.

**Launch template name and description**

Launch template name - *required*

Inventory-LT

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', "'", '@'.

Template version description

A prod webserver for MyApp

Max 255 chars

**Auto Scaling guidance** [Info](#)

Select this if you intend to use this template with EC2 Auto Scaling

☒ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

► Template tags

► Source template

**Launch template contents**

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

▼ **Application and OS Images (Amazon Machine Image) - required** [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Search our full catalog including 1000s of application and OS images

Recents | **My AMIs** | Quick Start

☒ Owned by me | ☐ Shared with me

[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

**Amazon Machine Image (AMI)**

Web Server AMI

ami-0dcaa417be0423a21

2023-10-28T16:12:50.000Z Virtualization: hvm ENA enabled: true Root device type: ebs

**Description**

Lab AMI for Web Server

**Architecture** x86\_64 **AMI ID** ami-0dcaa417be0423a21

**Summary**

**Software Image (AMI)**

Lab AMI for Web Server

ami-0dcaa417be0423a21

**Virtual server type (instance type)**

t2.micro

**Firewall (security group)**

Inventory-App

**Storage (volumes)**

1 volume(s) - 8 GiB

**Free tier:** In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 30 GiB of EBS storage, 2 million IOs, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Cancel **Create launch template**

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Module 9 Guided Lab - Creatin...Create launch template | EC2 |Target group details | EC2 | us-...EC2 | us-east-1

https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateTemplate:

awsServicesSearch[Alt+S]N. Virginiavoclabs/user2752985=103488117@student.swin.edu.au @ 2808-1373...

▼ Instance type InfoAdvanced

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Windows base pricing: 0.0162 USD per Hour

On-Demand SUSE base pricing: 0.0116 USD per Hour

On-Demand RHEL base pricing: 0.0716 USD per Hour

On-Demand Linux base pricing: 0.0116 USD per Hour

Free tier eligible

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

vockey

Create new key pair

▼ Network settings Info

Subnet Info

Don't include in launch template

Create new subnet

When you specify a subnet, a network interface is automatically added to your template.

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Select existing security group

Create security group

Security groups Info

Select security groups

Inventory-App sg-0be9535b41fed279b

VPC: vpc-045a22d2a5e361490

Compare security group rules

► Advanced network configuration

▼ Summary

Software Image (AMI)

Lab AMI for Web Server

ami-0dcaa417be0423a21

Virtual server type (instance type)

t2.micro

Firewall (security group)

Inventory-App

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 30 GiB of EBS storage, 2 million IOs, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Cancel

Create launch template



Name: Trac Duc Anh Luong - ID: 103488117

Module 9 Guided Lab - Creatin...

Create launch template | EC2 |

Target group details | EC2 | us-...

EC2 | us-east-1

https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateTemplate:

aws

Services

Search

[Alt+S]

N. Virginia

voclabs/user2752985=103488117@student.swin.edu.au @ 2808-1373...

▼ Advanced details Info

IAM instance profile Info

Inventory-App-Role

arn:aws:iam::280813733559:instance-profile/Inventory-App-Role

Create new IAM profile

Hostname type Info

Don't include in launch template

DNS Hostname Info

☐ Enable resource-based IPv4 (A record) DNS requests

☐ Enable resource-based IPv6 (AAAA record) DNS requests

Instance auto-recovery Info

Don't include in launch template

Shutdown behavior Info

Don't include in launch template

Not applicable for EC2 Auto Scaling

Stop - Hibernate behavior Info

Don't include in launch template

Not applicable for Amazon EC2 Auto Scaling.

Termination protection Info

Don't include in launch template

Stop protection Info

Don't include in launch template

Detailed CloudWatch monitoring Info

Enable

Additional charges apply

▼ Summary

Software Image (AMI)

Lab AMI for Web Server

ami-0dcaa417be0423a21

Virtual server type (instance type)

t2.micro

Firewall (security group)

Inventory-App

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 30 GiB of EBS storage, 2 million IOs, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Cancel

Create launch template

Name: Trac Duc Anh Luong - ID: 103488117

User data - optional

Info

Upload a file with your user data or enter it in the field.

Choose file

```
#!/bin/bash

Install Apache Web Server and PHP

yum install -y httpd mysql

amazon-linux-extras install -y php7.2

Download Lab files

wget https://aws-tc-largeobjects.s3-us-west-2.amazonaws.com/ILT-TF-200-ACACAD-20-EN/mod9-guided/scripts/inventory-app.zip

unzip inventory-app.zip -d /var/www/html/

Download and install the AWS SDK for PHP

wget https://github.com/aws/aws-sdk-php/releases/download/3.62.3/aws.zip

unzip aws -d /var/www/html

Turn on web server

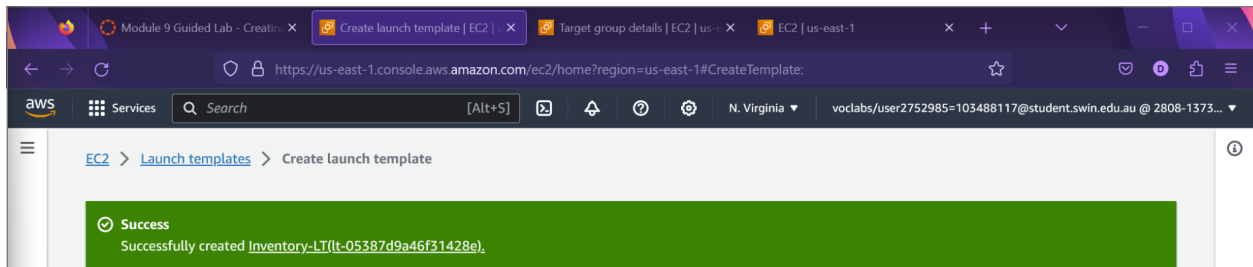
chkconfig httpd on

service httpd start
```

Cancel

Create launch template

☐ User data has already been base64 encoded



45. In the Success dialog, choose the **Inventory-LT** launch template.

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The screenshot shows the AWS Management Console interface for a Launch Template named 'Inventory-LT' (lt-05387d9a46f31428e) in the us-east-1 region. The left sidebar contains navigation links for EC2 Dashboard, EC2 Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Images, AMIs, AMI Catalog, Elastic Block Store, Volumes, Snapshots, Lifecycle Manager, Network & Security, Security Groups, Elastic IPs, Placement Groups, Key Pairs, Network Interfaces, and Load Balancing.

The main content area displays the 'Launch template details' for 'Inventory-LT'. The details include:

- Launch template ID: lt-05387d9a46f31428e
- Launch template name: Inventory-LT
- Default version: 1
- Owner: arn:aws:sts::280813733559:assumed-role/voclabs/user2752985=103488117@student.swin.edu.au

Below the details, there are tabs for 'Details', 'Versions', and 'Template tags'. The 'Details' tab is selected, showing 'Launch template version details' for version 1 (Default). The details include:

- Version: 1 (Default)
- Description: -
- Date created: 2023-10-28T16:19:46.000Z
- Created by: arn:aws:sts::280813733559:assumed-role/voclabs/user2752985=103488117@student.swin.edu.au

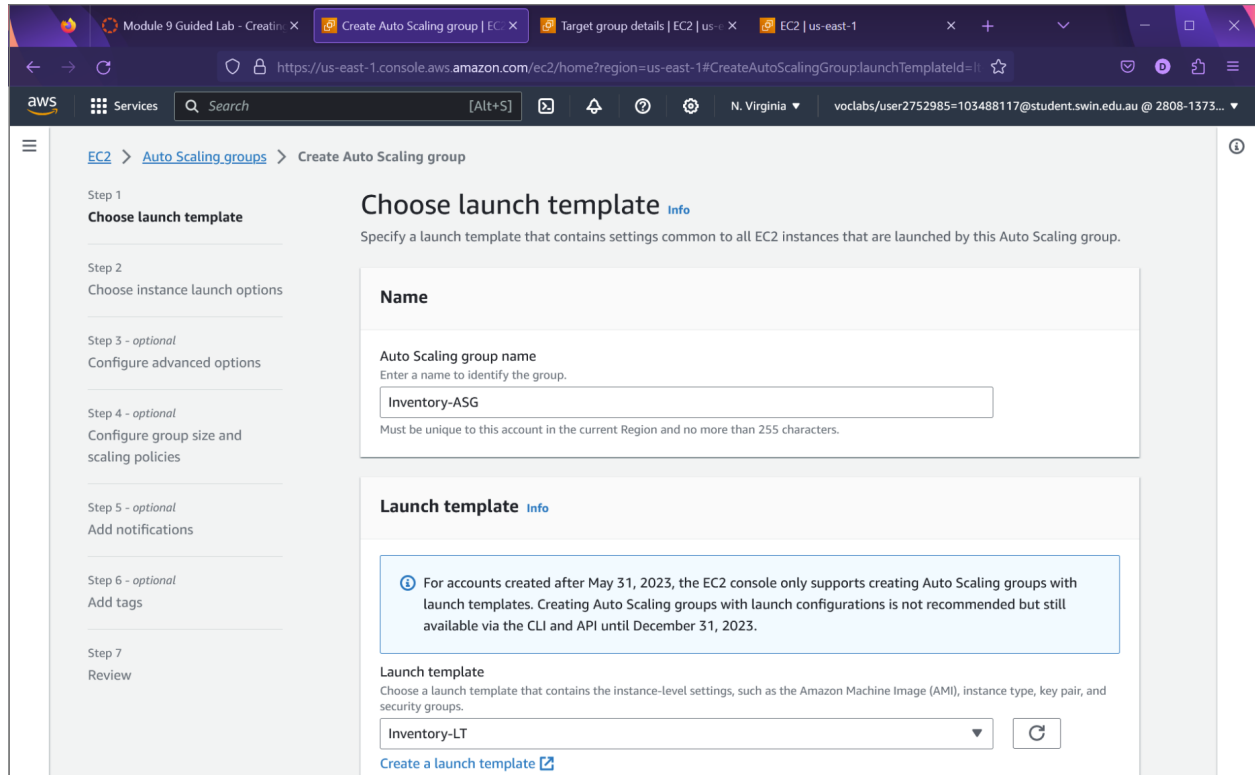
At the bottom, there are tabs for 'Instance details', 'Storage', 'Resource tags', 'Network interfaces', and 'Advanced details'. The 'Instance details' tab is selected, showing:

- AMI ID: ami-0dcaa417be0423a21
- Instance type: t2.micro
- Availability Zone: -
- Key pair name: vockey
- Security groups: -
- Security group IDs: sg-0be9535b41fed279b

46. From the Actions menu, choose *Create Auto Scaling group*

47. Configure the details in Step 1 (Choose launch template or configuration):

- **Auto Scaling group name:** **Inventory-ASG** (ASG stands for *Auto Scaling group*)
- **Launch template:** confirm that the *Inventory-LT* template you just created is selected.
- Choose Next



48. Configure the details in Step 2 (Choose instance launch options):

- **VPC:** choose *Lab VPC*
- **Availability Zones and subnets:** Choose *Private Subnet 1* and then choose *Private Subnet 2*. This will launch EC2 instances in private subnets across both Availability Zones.
- Choose Next

The screenshot shows the AWS Management Console interface for creating an Auto Scaling group. The browser tabs include 'Module 9 Guided Lab - Creating...', 'Create Auto Scaling group | EC2', 'Target group details | EC2 | us-east-1', and 'EC2 | us-east-1'. The URL is 'https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateAutoScalingGroup:launchTemplateId=...'. The page title is 'EC2 > Auto Scaling groups > Create Auto Scaling group'. The left sidebar shows a progress bar with steps: Step 1 (Choose launch template), Step 2 (Choose instance launch options), Step 3 - optional (Configure advanced options), Step 4 - optional (Configure group size and scaling policies), Step 5 - optional (Add notifications), Step 6 - optional (Add tags), and Step 7 (Review). The main content area is titled 'Choose instance launch options' with an 'Info' icon. Below the title is a description: 'Choose the VPC network environment that your instances are launched into, and customize the instance types and purchase options.' There are two main sections: 'Instance type requirements' and 'Network'. The 'Instance type requirements' section has an 'Override launch template' button and a table with columns 'Launch template', 'Version', and 'Description'. The table contains one row: 'Inventory-LT' (with a link icon), 'Default', and '-'. Below the table, it shows 'Instance type' as 't2.micro'. The 'Network' section has an 'Info' icon and a description: 'For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.' It includes a 'VPC' section with a dropdown menu showing 'vpc-045a22d2a5e361490 (Lab VPC)' and a 'Create a VPC' link. Below that is an 'Availability Zones and subnets' section with a dropdown menu showing 'Select Availability Zones and subnets' and a 'Create a subnet' link. The dropdown menu shows two selected subnets: 'us-east-1a | subnet-042cc62fb6f03286f (Private Subnet 1)' and 'us-east-1b | subnet-0b98dc1c036da94f5 (Private Subnet 2)'. At the bottom of the page are buttons: 'Cancel', 'Skip to review', 'Previous', and 'Next'.

Step 1  
[Choose launch template](#)

Step 2  
**Choose instance launch options**

Step 3 - optional  
[Configure advanced options](#)

Step 4 - optional  
[Configure group size and scaling policies](#)

Step 5 - optional  
[Add notifications](#)

Step 6 - optional  
[Add tags](#)

Step 7  
[Review](#)

## Choose instance launch options [Info](#)

Choose the VPC network environment that your instances are launched into, and customize the instance types and purchase options.

**Instance type requirements** [Info](#) [Override launch template](#)

You can keep the same instance attributes or instance type from your launch template, or you can choose to override the launch template by specifying different instance attributes or manually adding instance types.

| Launch template                                      | Version | Description |
|------------------------------------------------------|---------|-------------|
| <a href="#">Inventory-LT</a><br>lt-05387d9a46f31428e | Default | -           |

Instance type  
t2.micro

**Network** [Info](#)

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

**VPC**  
Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-045a22d2a5e361490 (Lab VPC)  
10.0.0.0/16

[Create a VPC](#)

**Availability Zones and subnets**  
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

us-east-1a | subnet-042cc62fb6f03286f (Private Subnet 1)  
10.0.2.0/23

us-east-1b | subnet-0b98dc1c036da94f5 (Private Subnet 2)  
10.0.4.0/23

[Create a subnet](#)

Cancel Skip to review Previous **Next**

49. Configure the details in Step 3 (Configure advanced options):

- In the **Load balancing** panel:
  - Choose **Attach to an existing load balancer**
  - **Existing load balancer target groups**: select *Inventory-App*.
- In the **Health checks** panel:
  - **Health check grace period**: 90 seconds
- In the **Additional settings** panel:
  - Select **Enable group metrics collection within CloudWatch**
- This will capture metrics at 1-minute intervals, which allows Auto Scaling to react quickly to changing usage patterns.
- Choose Next

The screenshot shows the AWS Management Console interface for creating an Auto Scaling group. The browser tabs include 'Module 9 Guided Lab - Creating...', 'Create Auto Scaling group | EC2', 'Target group details | EC2 | us-east-1', and 'EC2 | us-east-1'. The address bar shows the URL: <https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#CreateAutoScalingGroup:launchTemplateId=...>. The console header shows the AWS logo, 'Services', a search bar, and the region 'N. Virginia'. The user profile is 'voclabs/user2752985=103488117@student.swin.edu.au @ 2808-1373...'. The main navigation pane on the left shows the steps: Step 1: Choose launch template, Step 2: Choose instance launch options, Step 3 - optional: Configure advanced options (selected), Step 4 - optional: Configure group size and scaling policies, Step 5 - optional: Add notifications, Step 6 - optional: Add tags, and Step 7: Review. The main content area is titled 'Configure advanced options - optional' with an 'Info' icon. It includes a description: 'Integrate your Auto Scaling group with other services to distribute network traffic across multiple servers using a load balancer or to establish service-to-service communications using VPC Lattice. You can also set options that give you more control over health check replacements and monitoring.' The 'Load balancing' section has three options: 'No load balancer' (selected), 'Attach to an existing load balancer' (selected), and 'Attach to a new load balancer'. The 'Attach to an existing load balancer' section has two options: 'Choose from your load balancer target groups' (selected) and 'Choose from Classic Load Balancers'. The 'Existing load balancer target groups' section shows a dropdown menu with 'Select target groups' and a refresh button. A list of target groups is shown, including 'Inventory-App | HTTP' and 'Application Load Balancer: Inventory-LB'.

Step 1  
[Choose launch template](#)

Step 2  
[Choose instance launch options](#)

Step 3 - optional  
**Configure advanced options**

Step 4 - optional  
[Configure group size and scaling policies](#)

Step 5 - optional  
[Add notifications](#)

Step 6 - optional  
[Add tags](#)

Step 7  
[Review](#)

## Configure advanced options - optional [Info](#)

Integrate your Auto Scaling group with other services to distribute network traffic across multiple servers using a load balancer or to establish service-to-service communications using VPC Lattice. You can also set options that give you more control over health check replacements and monitoring.

### Load balancing [Info](#)

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

☐ No load balancer  
Traffic to your Auto Scaling group will not be fronted by a load balancer.

☒ Attach to an existing load balancer  
Choose from your existing load balancers.

☐ Attach to a new load balancer  
Quickly create a basic load balancer to attach to your Auto Scaling group.

### Attach to an existing load balancer

Select the load balancers that you want to attach to your Auto Scaling group.

☒ Choose from your load balancer target groups  
This option allows you to attach Application, Network, or Gateway Load Balancers.

☐ Choose from Classic Load Balancers

**Existing load balancer target groups**  
Only instance target groups that belong to the same VPC as your Auto Scaling group are available for selection.

Select target groups

Inventory-App | HTTP  
Application Load Balancer: Inventory-LB

**Health checks**  
Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.

EC2 health checks  
[Always enabled](#)

Additional health check types - optional [Info](#)

☐ Turn on Elastic Load Balancing health checks **Recommended**  
Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.

☐ Turn on VPC Lattice health checks  
VPC Lattice can monitor whether instances are available to handle requests. If it considers a target as failed a health check, EC2 Auto Scaling replaces it after its next periodic check.

Health check grace period [Info](#)  
This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.

90 seconds

**Additional settings**

Monitoring [Info](#)  
☒ Enable group metrics collection within CloudWatch

Default instance warmup [Info](#)  
The amount of time that CloudWatch metrics for new instances do not contribute to the group's aggregated instance metrics, as their usage data is not reliable yet.

☐ Enable default instance warmup

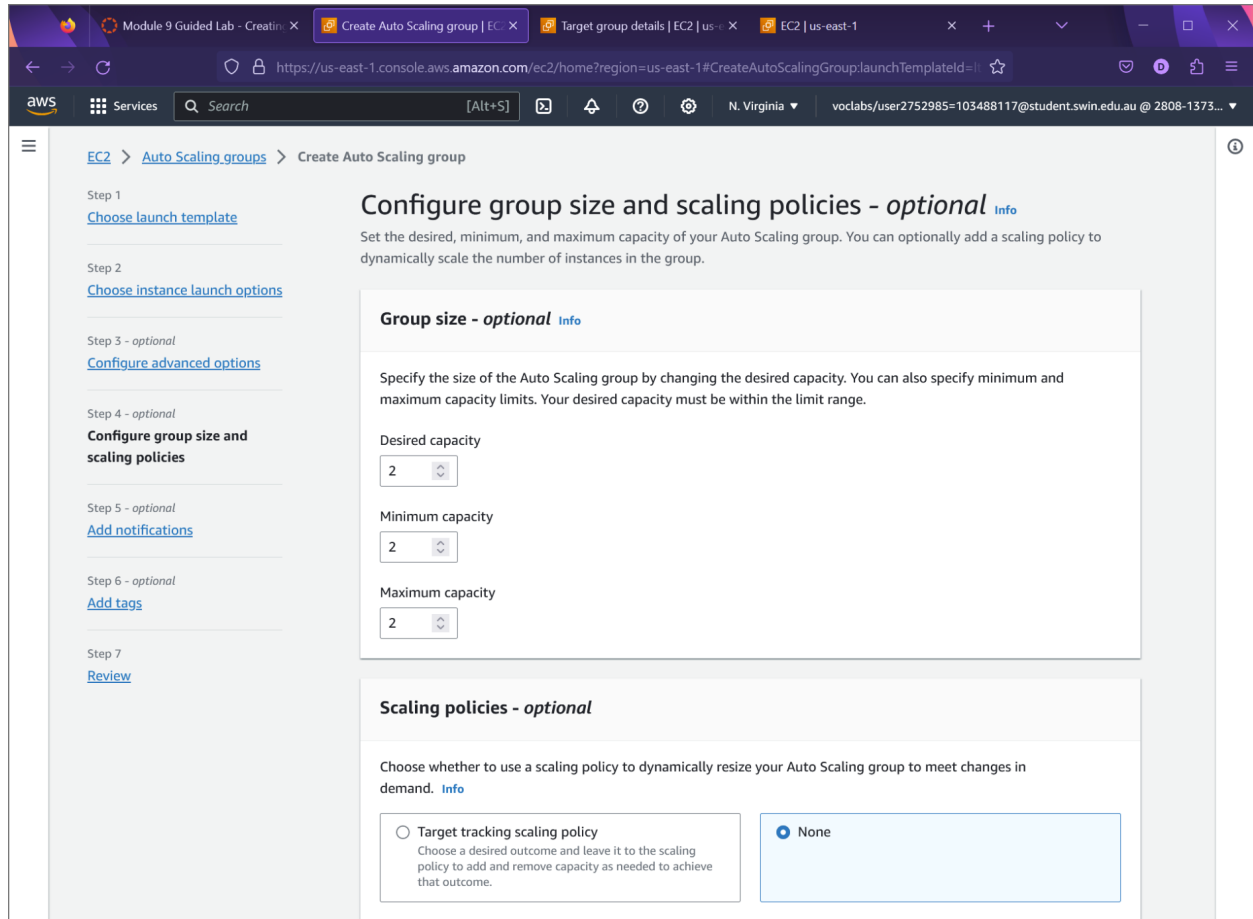
Cancel Skip to review Previous **Next**

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50. Configure the details in Step 4 (Configure group size and scaling policies - optional):

- Under **Group size**, configure:
  - **Desired capacity: 2**
  - **Minimum capacity: 2**
  - **Maximum capacity: 2**

This will allow Auto Scaling to automatically add/remove instances, always keeping between 2 and 6 instances running.
- Under **Scaling policies**, choose *None*  
For this lab, you will *maintain two instances at all times* to ensure *high availability*. If the application is expected to receive varying loads of traffic, you can also create *scaling policies* that define when to launch or terminate instances. However, you do not need to create scaling policies for the Inventory application in this lab.
- Choose Next



51. Configure the details in Step 5 (Add notifications - optional):

You do not need to configure any of these settings.

- Choose Next

52. Configure the details in Step 6 (Add tags - optional):

- Choose Add tag and Configure the following:

- **Key:** Name

- **Value:** Inventory-App

- These settings will *tag* the Auto Scaling group with a *Name*, which will also appear on the EC2 instances that are launched by the Auto Scaling group. You can use tags to identify which Amazon EC2 instances are associated with which application. You could also add tags such as *Cost Center* to assign application costs in the billing files.
- Choose Next



Step 1  
[Choose launch template](#)

Step 2  
[Choose instance launch options](#)

Step 3 - optional  
[Configure advanced options](#)

Step 4 - optional  
[Configure group size and scaling policies](#)

Step 5 - optional  
[Add notifications](#)

Step 6 - optional  
**Add tags**

Step 7  
Review

### Add tags - optional info

Add tags to help you search, filter, and track your Auto Scaling group across AWS. You can also choose to automatically add these tags to instances when they are launched.

ⓘ You can optionally choose to add tags to instances (and their attached EBS volumes) by specifying tags in your launch template. We recommend caution, however, because the tag values for instances from your launch template will be overridden if there are any duplicate keys specified for the Auto Scaling group.

#### Tags (1)

| Key                               | Value - optional                           | Tag new instances                   |                                       |
|-----------------------------------|--------------------------------------------|-------------------------------------|---------------------------------------|
| <input type="text" value="Name"/> | <input type="text" value="Inventory-App"/> | <input checked="" type="checkbox"/> | <input type="button" value="Remove"/> |

49 remaining

53. Configure the details in Step 7 (Review):

- Review the details of your Auto Scaling group
  - At the bottom of the page, choose Create Auto Scaling group
- Your Auto Scaling group will initially show an instance count of zero, but new instances will be launched to reach the **Desired** count of 2 instances.
- The *Inventory-ASG* will appear in the console

| Auto Scaling groups (1) <small>info</small> |               |                                |           |                   |                  |
|---------------------------------------------|---------------|--------------------------------|-----------|-------------------|------------------|
|                                             | Name          | Launch template/configuration  | Instances | Status            | Desired capacity |
| <input type="checkbox"/>                    | Inventory-ASG | Inventory-LT   Version Default | 0         | Updating capacity | 2                |

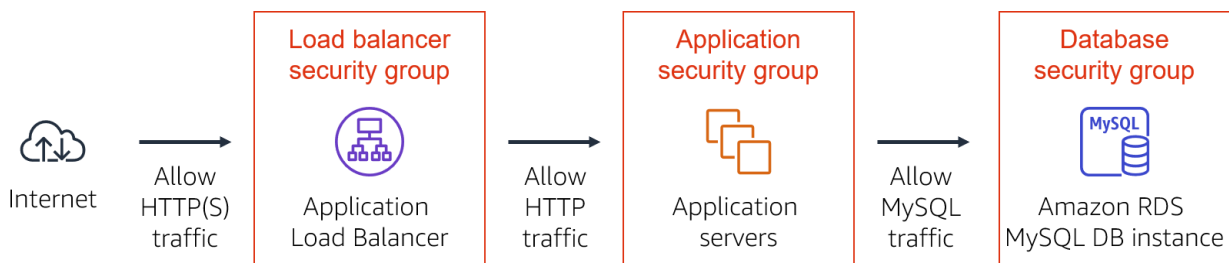
The review shows that:

- The group currently has *no instances*, but the **Status** column indicates *Updating capacity*.
- The **Desired** quantity is *2 instances*. Amazon EC2 Auto Scaling will attempt to launch two instances to reach the desired quantity.
- The **Min** and **Max** are also set to *2 instances*. Amazon EC2 Auto Scaling will try to always provide two instances, even if failure occurs.

- Your application will soon run across two Availability Zones. Amazon EC2 Auto Scaling will maintain that configuration even if an instance or Availability Zone fails.
- After a minute, choose the refresh icon to update the display. It should show that 2 *instances* are running.

## Task 4: Updating security groups

The application you deployed is a *three-tier architecture*. You will now configure the security groups to enforce these tiers:



### Load balancer security group

You already configured the *load balancer security group* when you created the load balancer. It accepts all incoming *HTTP* and *HTTPS* traffic.

The load balancer has been configured to forward incoming requests to a *target group*. When Auto Scaling launches new instances, it will automatically add those instances to the target group.

### Application security group

The *application security group* was provided as part of the lab setup. You will now configure it to only accept incoming traffic from the load balancer.

54. In the left navigation pane, choose **Security Groups**.

55. Select **Inventory-App**.

56. In the lower half of the page, choose the **Inbound rules** tab.

The security group is currently empty. You will now add a rule to accept incoming *HTTP* traffic from the load balancer. You do not need to configure *HTTPS* traffic because the load balancer was configured to forward HTTPS requests via HTTP. This practice offloads security to the load balancer, reducing the amount of work that is required by the individual application servers.

The screenshot displays the AWS Management Console interface for the 'us-east-1' region. The left-hand navigation pane shows the 'Network & Security' section expanded, with 'Security Groups' selected. The main content area shows a list of security groups under the heading 'Security Groups (1/5)'. The 'Inventory-App' security group is selected, and its details are shown in the lower half of the page. The 'Inbound rules' tab is active, showing a table with no rules. The 'Inventory-App' security group is associated with VPC ID 'vpc-045a22d2a5e361490' and has a description 'Enable access to the application servers'.

| Name          | Security group ID    | Security group name | VPC ID                | Description                              |
|---------------|----------------------|---------------------|-----------------------|------------------------------------------|
| -             | sg-01b5f67bc7cbc5fb7 | Inventory-LB        | vpc-045a22d2a5e361490 | Enable web access to the load balancer   |
| Inventory-App | sg-0be9535b41fed279b | Inventory-App       | vpc-045a22d2a5e361490 | Enable access to the application servers |
| -             | sg-070ce0feb7a63bc97 | default             | vpc-020fcee7b8223eee7 | default VPC security group               |
| -             | sg-0a132d0c92a8ff75d | default             | vpc-045a22d2a5e361490 | default VPC security group               |
| Inventory-DB  | sg-08944d5a8e7a7c783 | Inventory-DB        | vpc-045a22d2a5e361490 | Enable access to the database            |

**sg-0be9535b41fed279b - Inventory-App**

Details | **Inbound rules** | Outbound rules | Tags

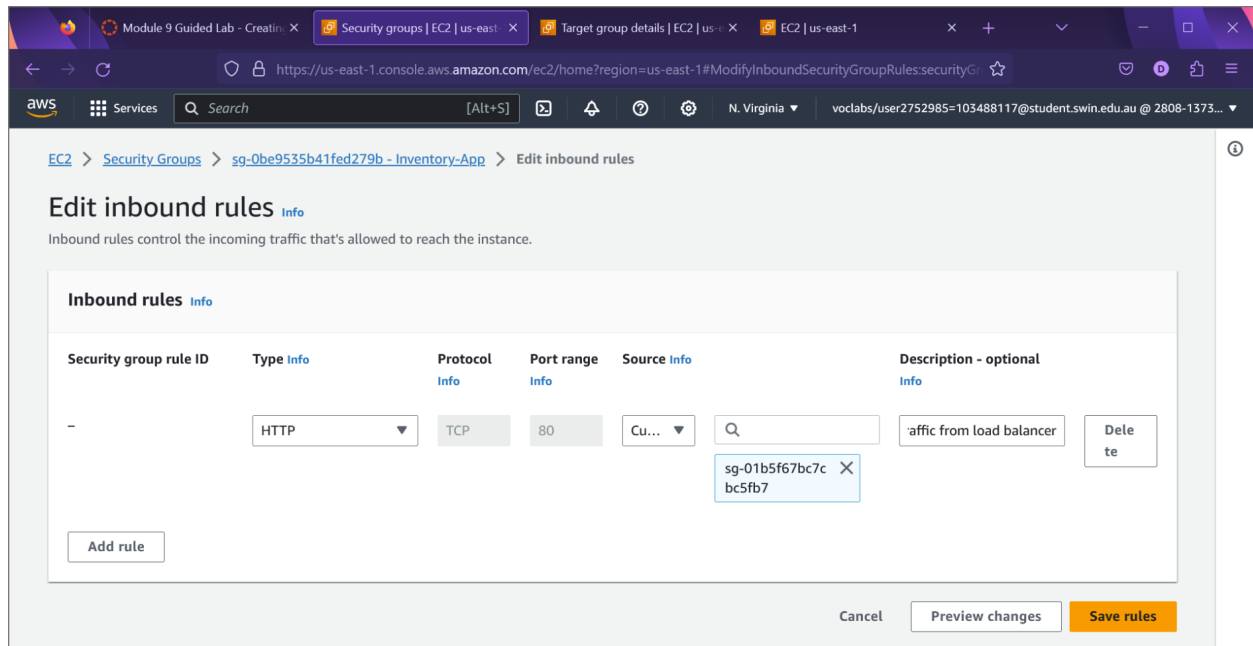
**Inbound rules**

| Name                          | Security group rule... | IP version | Type | Protocol |
|-------------------------------|------------------------|------------|------|----------|
| No security group rules found |                        |            |      |          |

57. Choose Edit inbound rules.

58. On the **Edit inbound rules** page, choose Add rule and configure these settings:

- **Type:** *HTTP*
    - **Source:**
      - Choose the search box to the right of **Custom**
      - Delete the current contents
      - Enter *sg*
      - From the list that appears, select **Inventory-LB**
  - **Description:** *Traffic from load balancer*
  - Choose Save rules
- The application servers can now receive traffic from the load balancer. This includes *health checks* that the load balancer performs automatically.



## Database security group

You will now configure the *database security group* to only accept incoming traffic from the application servers.

59. In the **Security groups** list, choose **Inventory-DB** (and make sure that no other security groups are selected).
- The existing rule permits traffic on port 3306 (used by MySQL) from any IP address within the VPC. This is a good rule, but security can be restricted further.

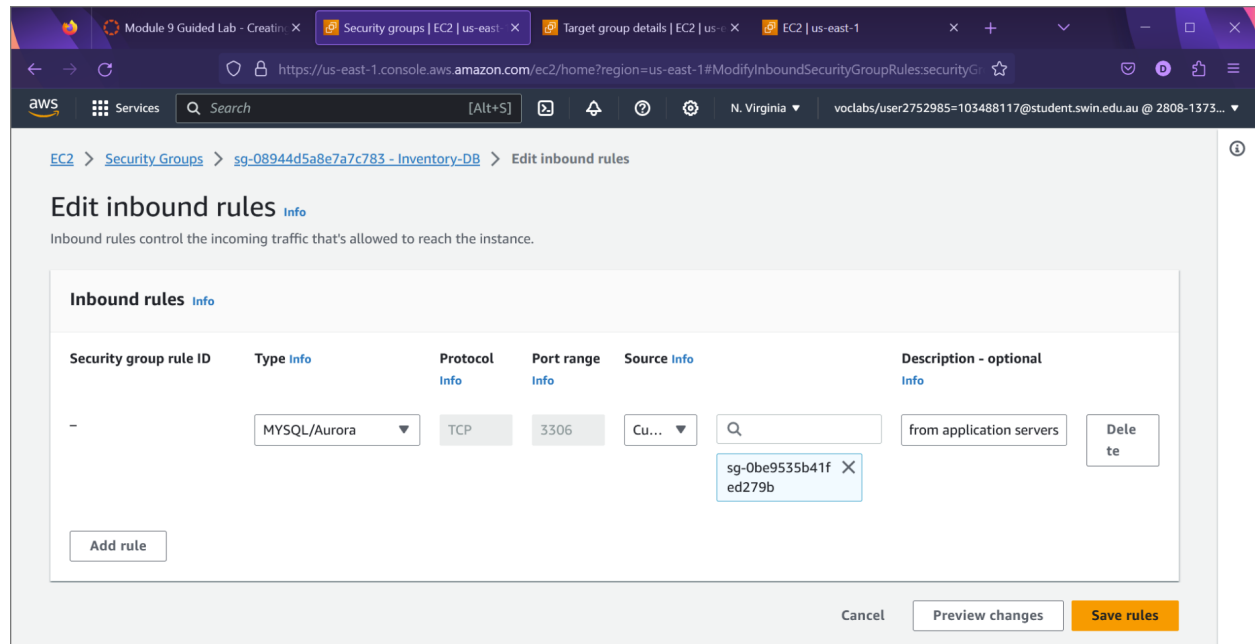
The screenshot shows the AWS Management Console interface. On the left is a navigation menu with categories like Instances, Images, Elastic Block Store, Network & Security, Load Balancing, and Auto Scaling. The 'Network & Security' section is expanded, showing 'Security Groups'. The main content area displays 'Security Groups (1/5)' with a table listing several groups. The 'Inventory-DB' group is selected. Below the table, the 'Inbound rules' tab is active, showing a single rule with the name 'sgr-061a33ffb1d848fd1', IP version 'IPv4', type 'MYSQL/Aurora', and protocol 'TCP'. Buttons for 'Manage tags' and 'Edit inbound rules' are visible.

| Name          | Security group ID    | Security group name | VPC ID                | Description       |
|---------------|----------------------|---------------------|-----------------------|-------------------|
| -             | sg-01b5f67bc7cbc5fb7 | Inventory-LB        | vpc-045a22d2a5e361490 | Enable web access |
| Inventory-App | sg-0be9535b41fed279b | Inventory-App       | vpc-045a22d2a5e361490 | Enable access to  |
| -             | sg-070ce0feb7a63bc97 | default             | vpc-020fceefb8223eee7 | default VPC sec   |
| -             | sg-0a132d0c92a8ff75d | default             | vpc-045a22d2a5e361490 | default VPC sec   |
| Inventory-DB  | sg-08944d5a8e7a7c783 | Inventory-DB        | vpc-045a22d2a5e361490 | Enable access to  |

| Name | Security group rule... | IP version | Type         | Protocol |
|------|------------------------|------------|--------------|----------|
| -    | sgr-061a33ffb1d848fd1  | IPv4       | MYSQL/Aurora | TCP      |

60. In the **Inbound rules** tab, choose Edit inbound rules and configure these settings:

- **Delete** the existing rule.
- Choose **Add rule**.
- For **Type**, choose **MYSQL/Aurora**
- Choose the search box to the right of **Custom**
- Enter **sg**
- From the list that appears, select **Inventory-App**
- **Description:** **Traffic from application servers**
- Choose Save rules



61. You have now configured *three-tier security*. Each element in the tier only accepts traffic from the tier above.

In addition, the use of private subnets means that you have two security barriers between the internet and your application resources. This architecture follows the best practice of applying multiple layers of security.

## Task 5: Testing the application

Your application is now ready for testing.

In this task, you will confirm that your web application is running. You will also test that it is highly available.

61. In the left navigation pane, choose **Target Groups**.

62. Select **Inventory-App**.

63. In the lower half of the page, choose the **Targets** tab.

This tab should show two *registered targets*. The **Health status** column shows the results of the load balancer health check that is performed against the instances.

64. In the **Registered targets** area, occasionally choose the refresh icon until the **Status** for both instances appears as *healthy*.

If the status does not eventually change to *healthy*, ask your educator for help with diagnosing the configuration.

You will test the application by connecting to the load balancer, which will then send your request to one of the EC2 instances. You will first need to retrieve the Domain Name System (DNS) name of the load balancer.

The screenshot shows the AWS Management Console interface. The left navigation pane is expanded, showing the 'Load Balancing' section with 'Target Groups' selected. The main content area displays the 'Target groups (1/1)' page for 'Inventory-App'. Below this, the 'Registered targets (2)' section is visible, showing two targets with 'Healthy' status.

| Instance ID     | Name          | Port | Zone       | Health status |
|-----------------|---------------|------|------------|---------------|
| i-07b265e5e1... | Inventory-App | 80   | us-east-1a | Healthy       |
| i-02d3827fd1... | Inventory-App | 80   | us-east-1b | Healthy       |

65. In the left navigation pane, choose **Load Balancers** and then choose **Inventory-LB**.

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66. In the **Details** tab in the lower half of the window, copy the **DNS name** to your clipboard.  
It should be similar to: *inventory-LB-xxxx.elb.amazonaws.com*

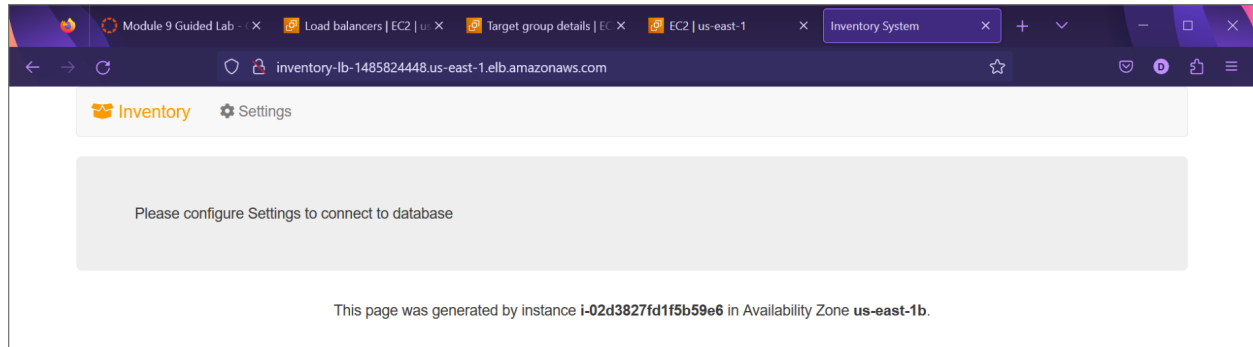
The screenshot shows the AWS Management Console interface. The left sidebar contains navigation links for EC2 Dashboard, EC2 Global View, Events, Instances, Images, Elastic Block Store, Network & Security, Load Balancing, and Auto Scaling. The main content area displays the 'Load balancers' list with a table showing one load balancer named 'Inventory-LB' in an 'Active' state. Below this, the 'Load balancer: Inventory-LB' details panel is open, showing the 'Details' tab. The details panel includes a table with the following information:

| Property           | Value                                                                                              |
|--------------------|----------------------------------------------------------------------------------------------------|
| Load balancer type | Application                                                                                        |
| Status             | Active                                                                                             |
| VPC                | vpc-045a22d2a5e361490                                                                              |
| IP address type    | IPv4                                                                                               |
| Scheme             | Hosted zone                                                                                        |
| Availability Zones | subnet-07d11ce7f2ed20b40 us-east-1b (use1-az6), subnet-07497f58e08ce1356 us-east-1a (use1-az4)     |
| Date created       | October 28, 2023, 23:10 (UTC+07:00)                                                                |
| Load balancer ARN  | arn:aws:elasticloadbalancing:us-east-1:280813733559:loadbalancer/app/inventory-LB/d0a93dff946f15b2 |
| DNS name           | Inventory-LB-1485824448.us-east-1.elb.amazonaws.com (A Record)                                     |

A tooltip indicates 'DNS name copied' for the DNS name field.

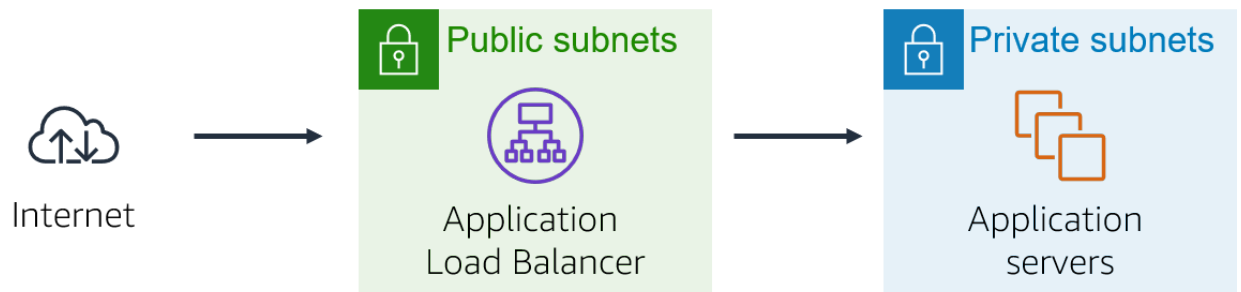
67. Open a new web browser tab, paste the DNS name from your clipboard and press ENTER.  
The load balancer forwarded your request to one of the EC2 instances. The instance ID and Availability Zone are shown at the bottom of the webpage.





68. Reload the page in your web browser. You should notice that the instance ID and Availability Zone sometimes toggles between the two instances.

When this web application displays, the flow of data over the network is:



- You sent the request to the *load balancer*, which resides in the *public subnets* that are connected to the internet.
- The load balancer chose one of the *EC2 instances* that reside in the *private subnets* and forwarded the request to it.
- The EC2 instance then returned the webpage content to the load balancer, which returned it to your web browser.

## Task 6: Testing high availability

Your application was configured to be highly available. You can prove the application's high availability by terminating one of the EC2 instances.

69. Return to the **EC2 console** tab in your web browser (but do not close the web application tab—you will return to it soon).

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70. In the left navigation pane, choose **Instances**.

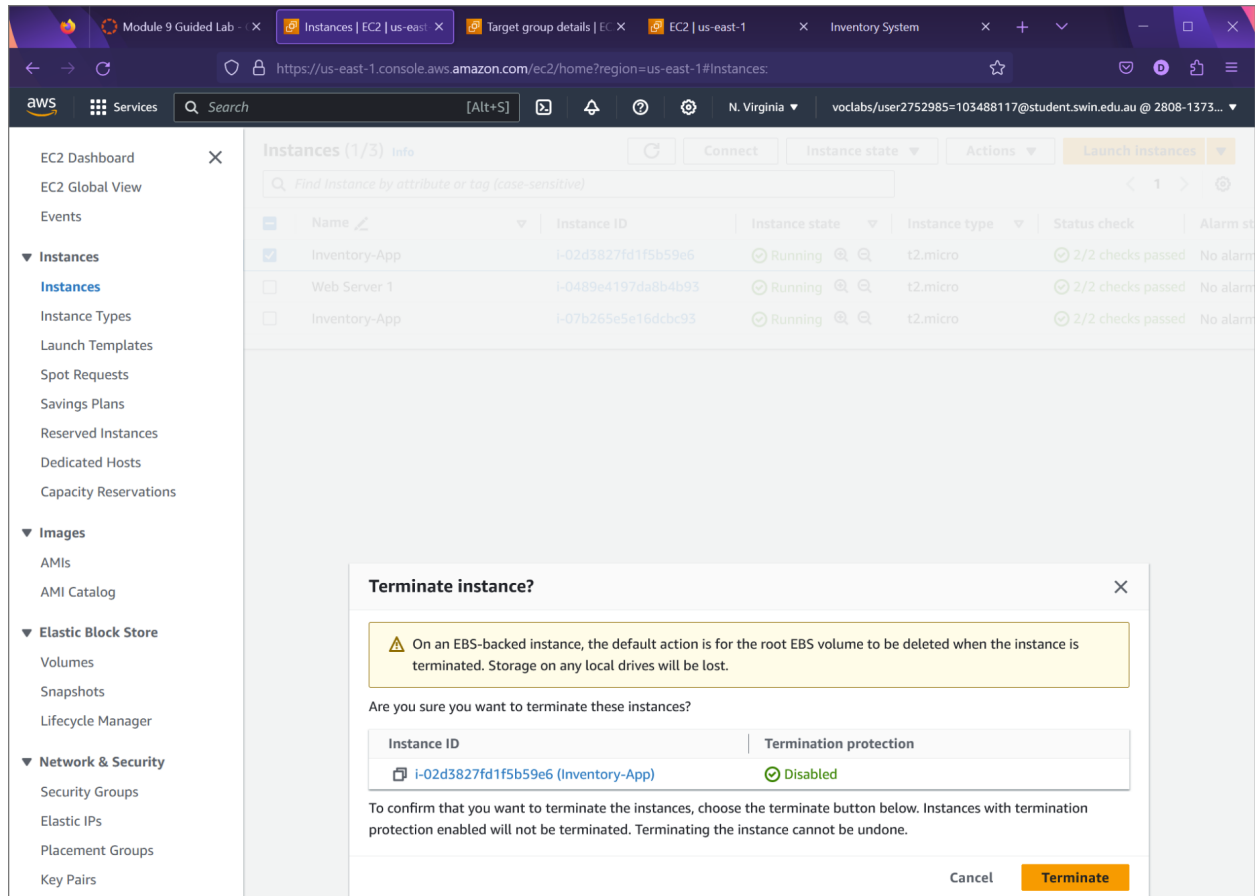
You will now terminate one of the web application instances to simulate a failure.

71. Select one of the **Inventory-App** instances (it does not matter which one you select).

72. Choose **Instance State > Terminate instance**.

73. Choose **Terminate**.

In a short time, the load balancer health checks will notice that the instance is not responding. The load balancer will automatically route all requests to the remaining instance.



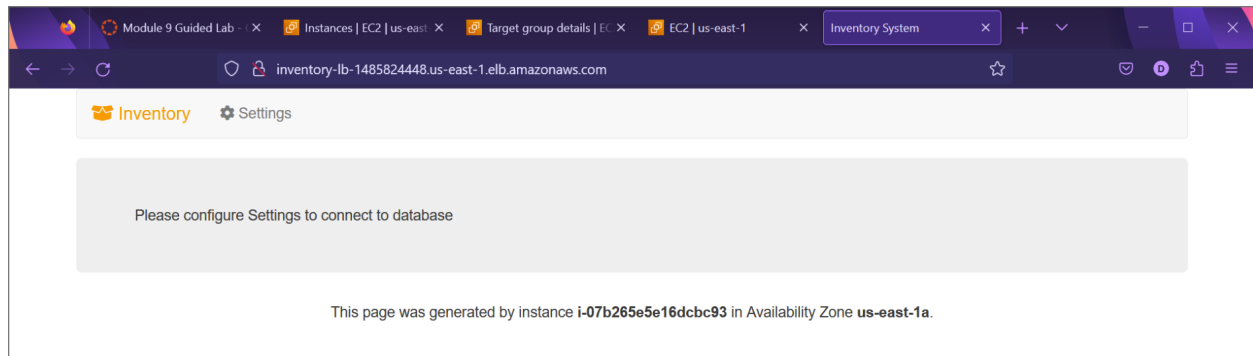
74. Return to the web application tab in your web browser and reload the page several times.

You should notice that the *Availability Zone* that is shown at the bottom of the page stays the same. Though an instance failed, your application remains available.

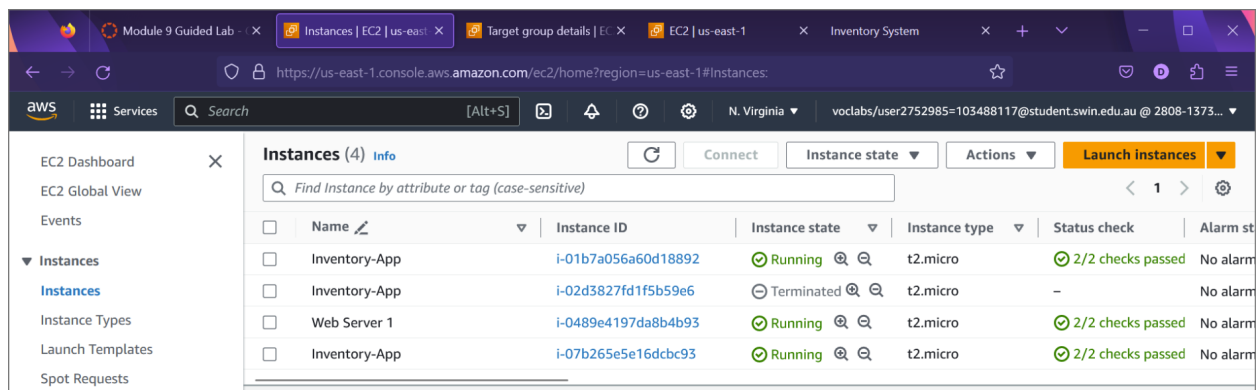
After a few minutes, Amazon EC2 Auto Scaling will also notice the instance failure. It was

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configured to keep two instances running, so Amazon EC2 Auto Scaling will *automatically launch a replacement instance*.



75. Return to the **EC2 console** tab where you have the instances list displayed. In the top-right area, choose the refresh icon every 30 seconds or so until a new EC2 instance appears. After a few minutes, the health check for the new instance should become healthy. The load balancer will resume sending traffic between the two Availability Zones. You can reload your web application tab to see this happen. This task demonstrates that your application is now *highly available*.

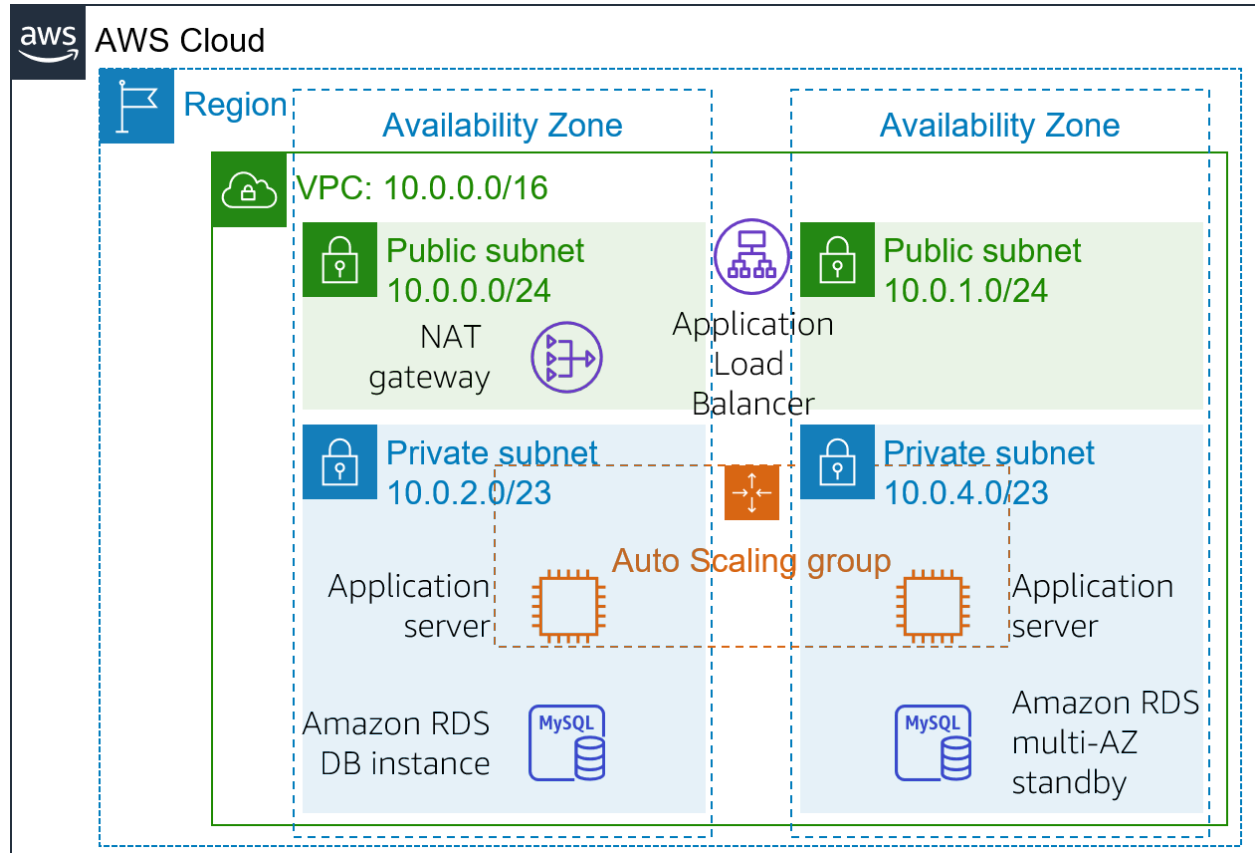


## Optional task 1: Making the database highly available

This task is *optional*. You can work on this task if you have remaining lab time.

The application architecture is now highly available. However, the Amazon RDS database operates from only one database instance.

In this optional task, you will make the database highly available by configuring it to run across multiple Availability Zones (that is, in a *Multi-AZ deployment*).



76. On the Services menu, choose **RDS**.

77. In the left navigation pane, choose **Databases**.

78. Choose the link for the name of the **inventory-db** instance.  
Feel free to explore the information about the database.

The screenshot shows the Amazon RDS console for the 'inventory-db' instance in the 'us-east-1' region. The instance is a MySQL Community Edition db.t3.micro class, currently 'Available' with 3.03% CPU usage and 0 connections. The 'Connectivity & security' tab is selected, showing the endpoint 'inventory-db.cr7utxjk1ykw.us-east-1.rds.amazonaws.com' on port 3306. It is connected to the 'us-east-1a' Availability Zone via the 'Lab VPC' (vpc-045a22d2a5e361490) and 'lab-db-subnet-group'. The security group is 'Inventory-DB' (sg-08944d5a8e7a7c783), which is active. The DB instance certificate expiration date is August 23, 2024, 00:08 (UTC+07:00). The 'Connected compute resources' section shows no resources are currently connected.

79. Choose Modify

80. Scroll down to the **Availability & durability** section. For **Multi-AZ deployment**, select **Create a standby instance**.

*Analysis:* You only need to reconfigure this one setting to convert the database to run across multiple data centers (Availability Zones).

This option does not mean that the database is *distributed* across multiple instances. Instead, one instance is the *primary* instance, which handles all requests. Another instance will be launched as the *standby* instance, which takes over if the primary instance fails. Your application continues to use the same DNS name for the database. However, the connections will automatically redirect to the currently active database server.

You can scale an EC2 instance by changing attributes, and you can also scale an RDS database this way. You will now scale up the database.

81. Scroll back up and for **DB instance class**, select **db.t3.small**.

This action doubles the size of the instance.

82. For **Allocated storage**, enter: **10**

This action doubles the amount of space that is allocated to the database.

Feel free to explore the other options on the page, but do not change any other settings.

The screenshot shows the AWS Management Console interface for configuring an Amazon RDS instance. The browser tabs at the top include 'Module 9 Guided Lab - Creatin...', 'RDS | us-east-1', and 'Inventory System'. The address bar shows the URL 'https://us-east-1.console.aws.amazon.com/rds/home?region=us-east-1#modify-instance:id=inventory-db'. The console header includes the AWS logo, 'Services', a search bar, and the user's profile information.

**Instance configuration**  
The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

▼ Hide filters

☐ Include previous generation classes

☐ Standard classes (includes m classes)

☐ Memory optimized classes (includes r and x classes)

☒ Burstable classes (includes t classes)

db.t3.small  
2 vCPUs 2 GiB RAM Network: 2,085 Mbps

**Storage**

Storage type [Info](#)

General Purpose SSD (gp2)  
Baseline performance determined by volume size

Allocated storage [Info](#)

10 GiB

The minimum value is 20 GiB and the maximum value is 16,384 GiB

ⓘ Provisioning less than 100 GiB of General Purpose (SSD) storage for high throughput workloads could result in higher latencies upon exhaustion of the initial General Purpose (SSD) IO credit balance. [Learn more](#)

ⓘ After you modify the storage for a DB instance, the status of the DB instance will be in storage-optimization. Your instance will remain available as the storage-optimization operation completes. [Learn more](#)

► Storage autoscaling

**Availability & durability**

Multi-AZ deployment [Info](#)

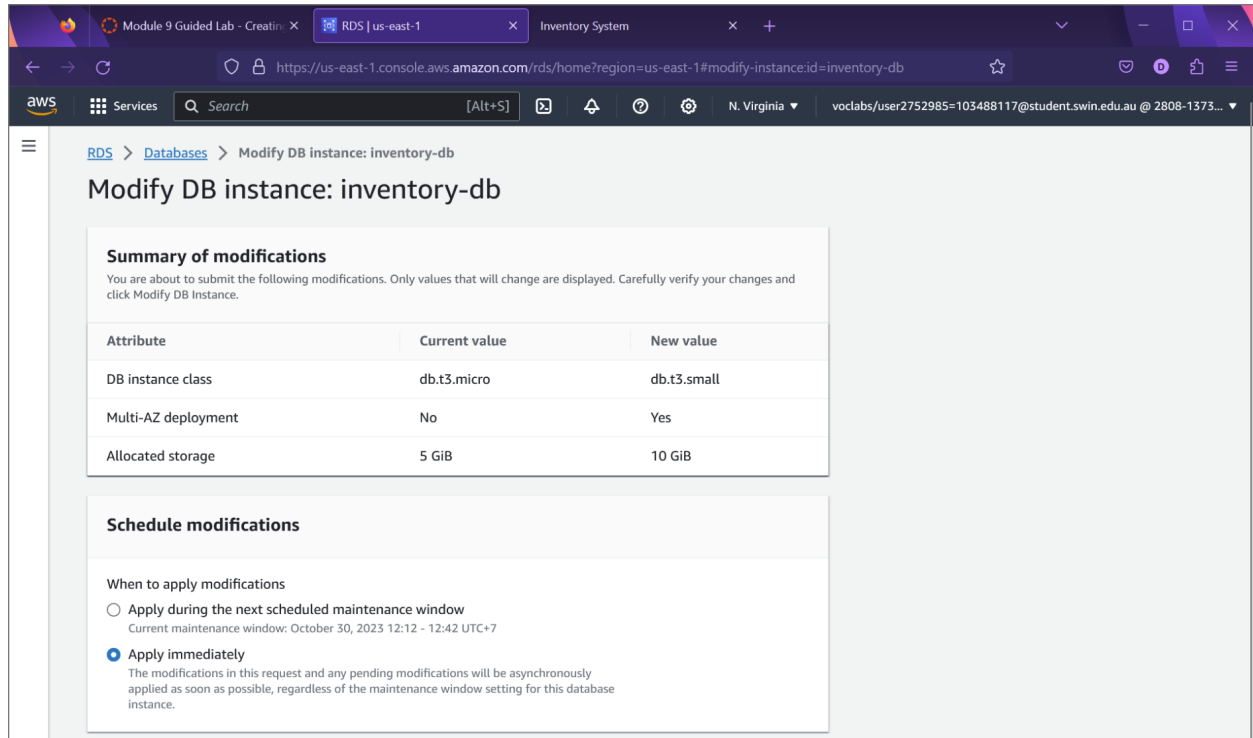
☐ Do not create a standby instance

☒ Create a standby instance (recommended for production usage)  
Creates a standby in a different Availability Zone (AZ) to provide data redundancy, eliminate I/O freezes, and minimize latency spikes during system backups.

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83. At the bottom of the page, choose Continue  
Database performance will be impacted by these changes. Therefore, these changes can be scheduled during a defined maintenance window, or they can be run immediately.

84. Under **Schedule modifications**, select **Apply immediately**.



85. Choose Modify DB instance

The database enters a *modifying* state while it applies the changes. You do not need to wait for it to complete.

## Optional task 2: Configuring a highly available NAT gateway

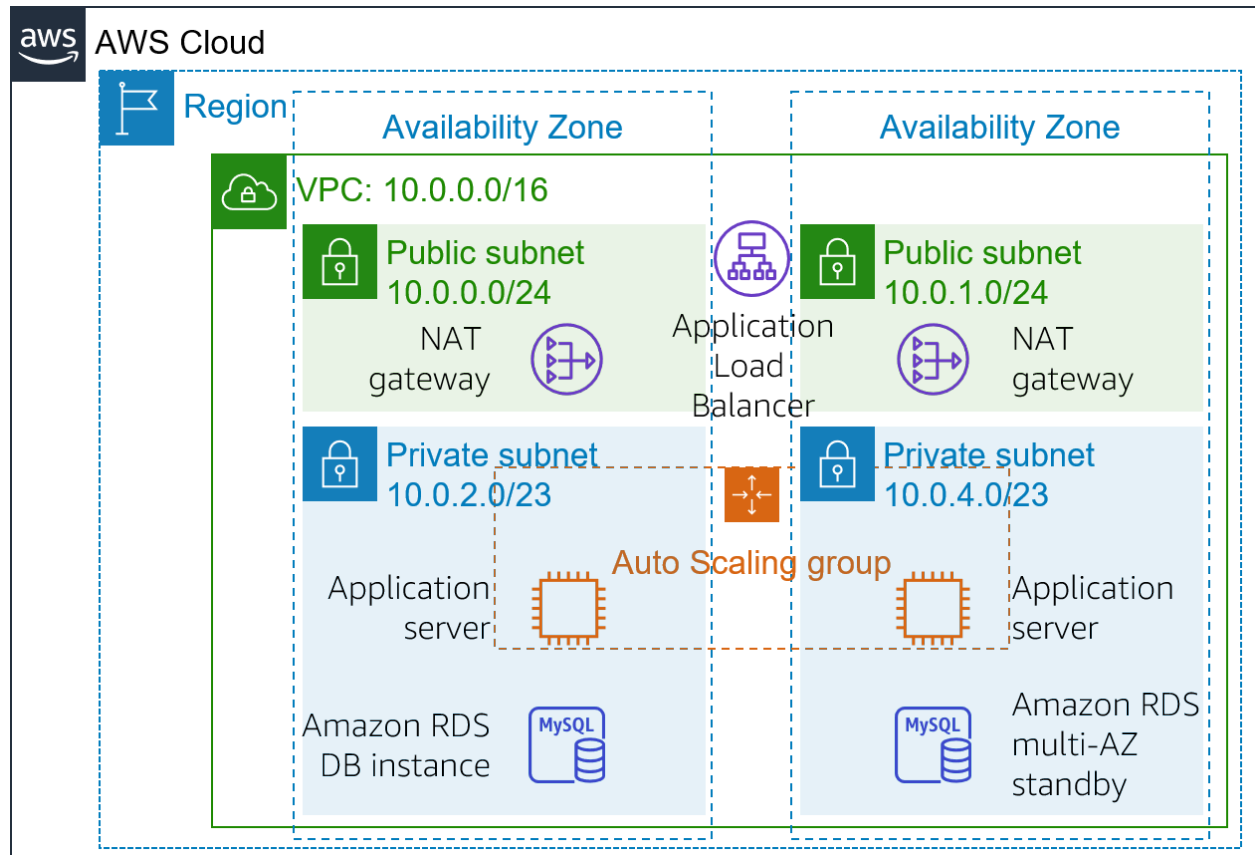
*This task is **optional**. You can work on this task if you have remaining lab time.*

The application servers run in a private subnet. If the servers must access the internet (for example, to download data), the requests must be redirected through a *Network Address Translation (NAT) gateway*. (The NAT gateway must be located in a public subnet).

The current architecture has only one NAT gateway in *Public Subnet 1*. Thus, if Availability Zone 1 fails, the application servers will not be able to communicate with the internet.

In this optional task, you will make the NAT gateway highly available by launching another NAT gateway in the other Availability Zone. The resulting architecture will be highly available:





86. On the Services menu, choose **VPC**.

87. In the left navigation pane, choose **NAT gateways**.

The existing NAT gateway displays. You will now create a NAT gateway for the other Availability Zone.

88. Choose Create NAT gateway and configure these settings:

- **Subnet:** *Public Subnet 2*
- Choose Allocate Elastic IP
- Choose Create NAT gateway

You will now create a new route table for *Private Subnet 2*. This route table will redirect traffic to the new NAT gateway.

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https://us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#CreateNatGateway:

aws Services Search [Alt+S] N. Virginia voclabs/user2752985=103488117@student.swin.edu.au @ 2808-1373...

✓ Elastic IP address 34.204.16.134 (eipalloc-09604d07d076e21fe) allocated.

VPC > NAT gateways > Create NAT gateway

### Create NAT gateway [Info](#)

A highly available, managed Network Address Translation (NAT) service that instances in private subnets can use to connect to services in other VPCs, on-premises networks, or the internet.

#### NAT gateway settings

**Name - optional**  
Create a tag with a key of 'Name' and a value that you specify.

The name can be up to 256 characters long.

**Subnet**  
Select a subnet in which to create the NAT gateway.

subnet-07d11ce7f2ed20b40 (Public Subnet 2)

**Connectivity type**  
Select a connectivity type for the NAT gateway.

☒ Public  
☐ Private

**Elastic IP allocation ID [Info](#)**  
Assign an Elastic IP address to the NAT gateway.

eipalloc-09604d07d076e21fe [Allocate Elastic IP](#)

▶ **Additional settings** [Info](#)

#### Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add new tag](#)

You can add 50 more tags.

[Cancel](#) [Create NAT gateway](#)

89. In the left navigation pane, choose **Route tables**.

90. Choose Create route table and configure these settings:

- **Name:** Private Route Table 2
- **VPC:** Lab VPC
- Choose **Create route table**.

[VPC](#) > [Route tables](#) > Create route table

## Create route table [Info](#)

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

### Route table settings

**Name - optional**  
Create a tag with a key of 'Name' and a value that you specify.

**VPC**  
The VPC to use for this route table.

### Tags

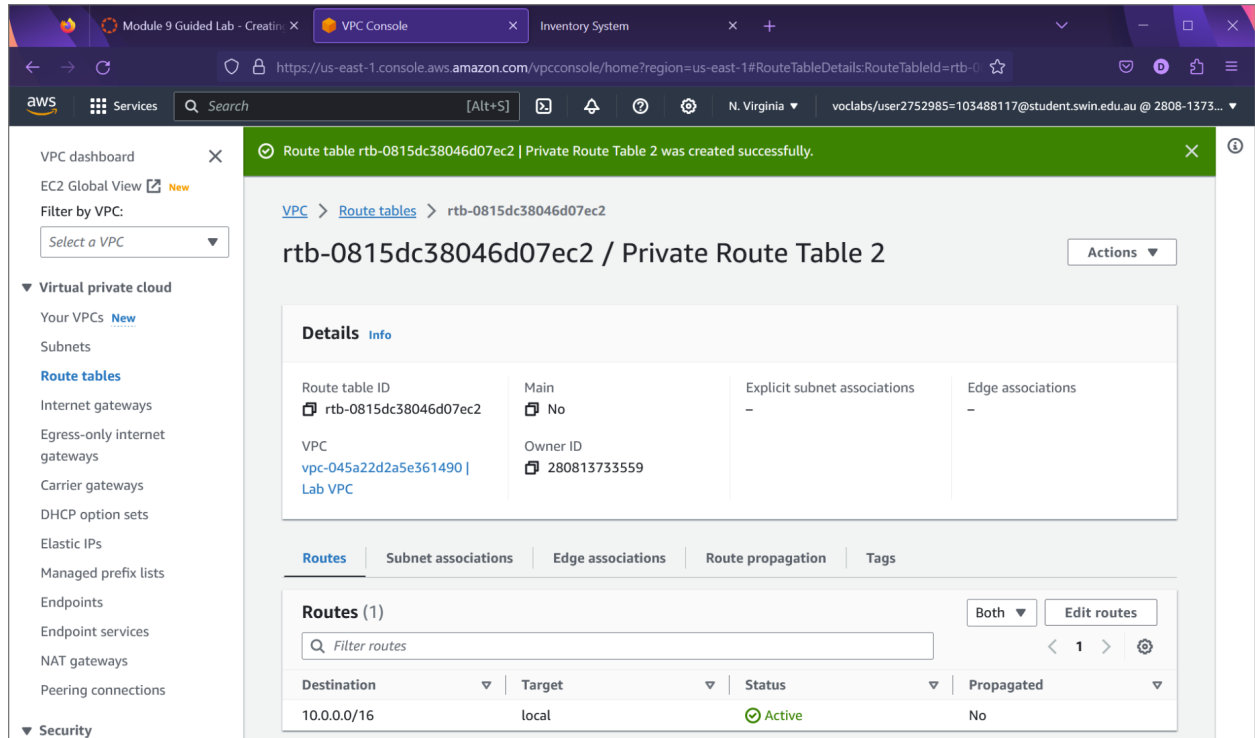
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

| Key                               | Value - optional                                   |                                       |
|-----------------------------------|----------------------------------------------------|---------------------------------------|
| <input type="text" value="Name"/> | <input type="text" value="Private Route Table 2"/> | <input type="button" value="Remove"/> |

You can add 49 more tags.

91. Observe the settings in the **Routes** tab.

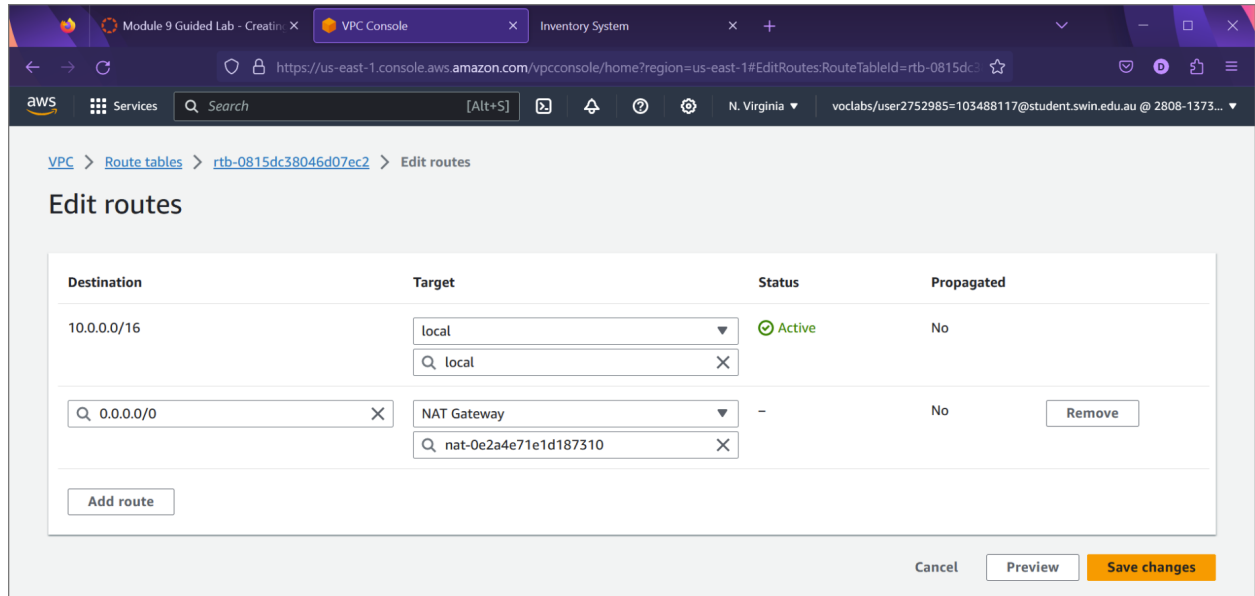
Currently, one route directs all traffic *locally*. You will now add a route to send internet-bound traffic through the new NAT gateway.



92. Choose Edit routes and then configure these settings:

- Choose Add route
- **Destination:** 0.0.0.0/0
- **Target:** Select *NAT Gateway*, then select the *nat-* entry that is *not* the entry for *\_NATGateway1*  
**Tip:** To discover which *nat-* entry is *NOT* the one to select, choose the **Details** button above these instructions and then select **Show**. the name of *NATGateway1* is displayed. Return to the VPC console and select the Target that is *NOT* *NATGateway1*.
- Choose **Save changes**

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93. Choose the **Subnet associations** tab.

94. Choose Edit subnet associations

95. Select **Private Subnet 2**.

Updated routes for rtb-0815dc38046d07ec2 / Private Route Table 2 successfully

Details

VPC > Route tables > rtb-0815dc38046d07ec2 > Edit subnet associations

### Edit subnet associations

Change which subnets are associated with this route table.

**Available subnets (1/4)**

Filter subnet associations

|                                     | Name             | Subnet ID                                | IPv4 CIDR   | IPv6 CIDR | Route table ID                                 |
|-------------------------------------|------------------|------------------------------------------|-------------|-----------|------------------------------------------------|
| <input type="checkbox"/>            | Public Subnet 2  | <a href="#">subnet-07d11ce7f2ed20b40</a> | 10.0.1.0/24 | –         | <a href="#">rtb-09be1677cf8a63dc0 / Public</a> |
| <input type="checkbox"/>            | Public Subnet 1  | <a href="#">subnet-07497f58e08ce1356</a> | 10.0.0.0/24 | –         | <a href="#">rtb-09be1677cf8a63dc0 / Public</a> |
| <input checked="" type="checkbox"/> | Private Subnet 2 | <a href="#">subnet-0b98dc1c036da94f5</a> | 10.0.4.0/23 | –         | <a href="#">rtb-0815dc38046d07ec2 / Privat</a> |
| <input type="checkbox"/>            | Private Subnet 1 | <a href="#">subnet-042cc62fb6f03286f</a> | 10.0.2.0/23 | –         | <a href="#">rtb-0958b3065da123923 / Privat</a> |

**Selected subnets**

subnet-0b98dc1c036da94f5 / Private Subnet 2

Cancel Save associations

96. Choose Save associations

This action now sends internet-bound traffic from Private Subnet 2 to the NAT gateway that is in the same Availability Zone.

Your NAT gateways are now highly available. A failure in one Availability Zone will not impact traffic in the other Availability Zone.